

TruAwake Application



SUBMITTED BY

ANEESHA KHAN (FSD-SP-062)

RABIA RAFIQ (FSD-SP-084)

SUPERVISED BY

MUHAMMAD UMAR

CO-SUPERVISED BY

M. AHSAN AFTAB

*Submitted for the partial fulfillment of BS software Engineering Degree to
the Faculty of SE & CS*

NATIONAL UNIVERSITY OF MODERN LANGUAGES

FAISALABAD

December, 2025

CERTIFICATE

Dated: _____

Final Approval

It is certified that the project report title ‘**TruAwake-AI based driver drowsiness detection application using Real-Time Image Processing**’ submitted by Aneesha Khan and Rabia Rafiq for the partial fulfillment of the requirement of “**Bachelor’s Degree in Software Engineering**” is approved.

COMMITTEE

Dr. Muhammad Tahir

HoD Engineering

Signature: _____

Mr. Rao Umar Farooq

Head Project Committee

Signature: _____

Mr. Muhammad Umar

Supervisor

Signature: _____

DECLARATION

All undersigned authors state that the Final Year Project (FYP) documentation includes our original work from undergraduate studies. We confirm that the contents of this project are entirely original and have no unauthorized source plagiarism which is combined with proper citation of all references and acknowledgements.

The policies defined by our university will take effect when academic dishonesty or plagiarism occurs which could lead to project failure and additional disciplinary measures. The work we presents maintain its genuine quality because it complies with complete integrity and all institutional guidelines.

Group Members

Signature

Aneesha Khan

Rabia Rafiq

PLAIGRISM CERTIFICATE

This is to certify that the project entitled “**TruAwake Application**(driver drowsiness detection application)”, is being submitted here for the award of the “**Degree of Bachelor**” in “**Software Engineering**”. This is the result of the original work by **Aneesha Khan** and **Rabia Rafiq** under my supervision and guidance. The work embodied in this project has not been done earlier for the basis of the award of any degree or compatible certificate or similar title of this for any other diploma/examining body or university to the best of my knowledge and belief.

Turnitin Originality Report

Processed on 02-Dec-2025 08:42AM ID: 3430502726

Word Count: 15,568

Similarity Index 9% Similarity by Source Internet Sources: 6%

Publications: 3%

Student Papers: 7%

Muhammad Umar

ACKNOWLEDGEMENT

We are grateful to our supervisor who helped us a lot with our project. Our supervisor **Mr. Muhammad Umar** helped us to stay on track throughout the project and helped us in our project. We are grateful for the time you took to provide us with insightful feedback, suggestions, and advice. Your expertise and knowledge helped us to navigate the challenges we encountered along the way and provided us with a better understanding of our project.

ABSTRACT

TruAwake is a driver drowsiness detection application, designed to magnify road and human life safety by averting drowsiness and fatigue-related accidents. Taking advantage of AI-driven image processing and mobile sensors, the application scans the facial expressions, namely eye closure, yawning, head tilts and body movements to ascertain the rapid signs of drowsiness and fatigue. Far apart from regular drowsiness detection systems that depend on expensive hardware and intrusive wearable devices that is not much easier for drivers to use, TruAwake impart a non-intrusive, cost-effective proxy that makes use of smartphone cameras by ensuring the accessibility and ease of use. This system will provide real-time alerts, adaptive pattern learning and driving behavior analysis to increase accuracy. By ensuring the users privacy and security, app will process the data directly on the devices countries all over the world dealing with road accidents commence due to drowsiness and fatigue, Pakistan also experiencing the high rate of drowsiness and fatigue related accidents on motorways and highways specially in summer and in December, TruAwake put on a critical need for a non- intrusive, reliable , mobile-based detection system.

List of Abbreviations

Abbreviation	Full Form
PERCLOS	Percentage of Eye Closure
GPS	Global Positioning System
Iot	Internet of Things
API	Application Programming Interface
NLP	Natural Language Processing
DMS	Driver Monitoring System
EA	Environmental Adaptability
CE	Cost Effectiveness
SSA	Sensors & Signal Analysis
SSL	Safet Signal Level
CDM	Combined Detection Method
WHO	World Health Organization
JWT	JSON Web Token
RAM	Random Access Memory
SDG	Sustainable Development Goals
XML	Extensible Markup Language
FPS	Frames Per Second
SDK	Software Development Kit
RAM	Random Access Memory
OCR	Optical Character Recognition
ROI	Region of Interest

Table of Contents

CHAPTER 1.....	1
1.1 Introduction.....	2
1.1.1 Problem Domain	2
1.1.2 Proposed System.....	2
1.2 Motivation.....	2
1.3 Problem Statement.....	3
1.4 Aims and Objectives	3
1.5 Scope of the Study	3
1.6 Process Model	4
1.6.1 User registration and Authentication.....	4
1.6.2 Data collection	4
1.6.3 AI-driven Analysis.....	4
1.6.4 Alert System	4
1.6.5 Post-driving report	4
1.7 Proposed System Features.....	5
1.8 Development Methodology.....	5
1.9 Implementation	6
1.9.1 Technology Stack.....	6
1.9.2 Frontend Development.....	6
1.9.3 Backend Development	6
1.9.4 Testing & Debugging	6
1.9.5 Deployment & Maintenance	6
1.10 Nature of Project.....	6
1.11 Overview.....	7
CHAPTER 2.....	8
2.1 Introduction.....	9
2.1.1 Background.....	9
2.2 Types of Existing Systems.....	10
2.2.1 Physiological Monitoring System	10
2.2.2 Behavioral Analysis Systems	12
2.2.3 Vehicle-Based Detection Systems	13

2.3 Existing Systems/Related Work	15
2.4 Comparison of Past Papers with our Project Idea.....	16
2.5 Explanation of Important Constructs of the Application.....	17
2.5.1 Domain	17
2.5.2 Drowsiness Detection	18
2.5.3 Mobile Sensor Integration.....	18
2.5.4 Image Processing	18
2.5.5. Real-Time Alert Mechanism.....	18
2.5.6 Adaptive Learning	18
2.5.7 Location-Based Sensitivity	19
2.5.8 Post-Driving Reporting.....	19
2.5.9 Privacy and On-Device Processing.....	19
2.6 Summary.....	19
CHAPTER 3.....	21
3.1 Introduction.....	22
3.2 Interface Requirements	22
3.2.1 Hardware Interface Requirements.....	22
3.2.2 Software Interface Requirements.....	23
3.3 Functional Requirements.....	24
3.3.1 User Authentication.....	24
3.3.2 Real-Time Facial Monitoring.....	24
3.3.3 Drowsiness Detection Algorithm.....	24
3.3.4 Sensor Data Collection.....	24
3.3.5 Alert System	25
3.3.6 Location-Based Alert Sensitivity	25
3.3.7 Post Driving Report	25
3.3.8 Data Privacy.....	25
3.4 Use Case Model Process	25
3.4.1 Actors:.....	25
3.5 Use Cases.....	28
3.5.1 Use Case Description	28
3.6 Non-Functional Requirements	31
3.6.1 Performance	31

3.6.2 Reliability.....	32
3.6.3 Security.....	32
3.6.4 Consistency.....	32
3.6.5 Usability.....	32
3.6.6 Maintainability.....	33
3.6.7 Portability.....	33
3.7 Requirement Resources	33
3.7.1 Equipment.....	33
3.7.2 Software Tools and Technologies.....	34
3.7.3 Funds.....	34
3.7.4 Human Efforts	34
3.7.5 Database Requirements.....	35
3.8 Project Feasibility	36
3.8.1 Technical Feasibility.....	36
3.8.2 Operational Feasibility.....	37
3.8.3 Legal and Ethical Feasibility	37
3.9 Summary.....	38
CHAPTER 4.....	39
4.1 Introduction.....	40
4.2 System Design	40
4.2.1 Client Side(Mobile App)	40
4.2.2 Hardware Architecture.....	41
4.2.3 System Modules.....	41
4.2.4 Limitations	43
4.2.5 SDG Impact	43
4.2.6 Technology Stack.....	43
4.2.7 Applications & Impact.....	44
4.3 Design Approach	44
4.3.1 Top-Down Design Approach.....	44
4.3.2 Bottom-Up Design Approach	44
4.3.3 Design Approach Used in TruAwake Application	45
4.4 Interface Design.....	45
4.4.1 High Fidelity Prototype	45

4.5 4+1 View Model of Architecture	56
4.5.1 Logical View.....	56
4.5.2 Process View.....	56
4.5.3 Development View	59
4.5.4 Physical View	60
4.6 Entity Relationship Diagram	61
4.7 Summary.....	62
CHAPTER 5.....	63
5.1 Introduction.....	64
5.2 Modules of TruAwake	64
5.2.1 User Authentication	64
5.2.2 Alarm Module.....	64
5.2.3 Analytics and Report Module	64
5.2.4 Theme and UI Customization Module.....	65
5.2.5 Notifications and Reminders Module	65
5.3 Hardware Modules.....	65
5.3.1 Android Smartphone/Tablet.....	65
5.3.2 2GB RAM(Minimum)	65
5.3.3 300MB Free Storage	65
5.3.4 Mobile Holder or Mount.....	65
5.4 Summary.....	66
CHAPTER 6.....	67
6.1 Introduction.....	68
6.2 Test/ Validate Achievements	68
6.2.1 Test-Case 1: Camera Activation & Real-Time Fee	68
6.2.2 Test-Case 2: Facial Landmark Detection.....	69
6.2.3 Test-Case 3: Drowsiness Detection Accuracy	71
6.2.4 Test-Case 4: Alert Mechanism.....	72
6.2.5 Test-Case 5: History Recording.....	73
6.2.6 Test-Case 6: Search Features	74
6.3 Analysis	75
6.4 Validation.....	75
6.5 Testing	75

6.6 Summary 75

CHAPTER 7..... 77

7.1 Introduction..... 78

7.2 Project Review and Achievements 78

7.3 Limitations and Claims 78

7.4 Future Work and Recommendations..... 79

7.5 Summary 79

REFERENCES..... 81

List of Figures

Figure 1.1 Working Flow of TruAwake.....	5
Figure 2.1 Driver Drowsiness Measures	10
Figure 2.2 Driver Drowsiness Detection Using Behavioral Measures.....	12
Figure 2.3 Smart Eye Pro and EEG signals with dynamical encoder and decoder.	16
Figure 3.1 Use Case Model of TruAwake.....	27
Figure 3.2 User Login: Access to Dashboard.....	28
Figure 3.3 Driver Drowsiness Detection Pattern.....	29
Figure 3.4 Alert Generation When Drowsiness Detected	30
Figure 3.5 Post-Driving Report.....	30
Figure 3.6 Driver Account Management(Optional)	31
Figure 4.1 Hardware Architecture of TruAwake	41
Figure 4.2 Data Flow Diagram (Level 0).....	42
Figure 4.3 Create Account, Sign in.	46
Figure 4.4 My Profile.....	47
Figure 4.5 Vehicle Registration Screen.....	48
Figure 4.6 License Verification.	49
Figure 4.7 Home/Dashboard Screen	49
Figure 4.8 Screen Recording/Start Detection.....	50
Figure 4.9 Location Screen.	51
Figure 4.10 Post Driving Report.	52
Figure 4.11 Notification Screen.	52
Figure 4.12 Historical Data/Post Driving Report.	53
Figure 4.13 Activity Screens.	54
Figure 4.14 Profile Updation Screen.	55
Figure 4.15 Theme.	55

Figure 4.16 Settings.....	55
Figure 4.17 Power & Performance.....	55
Figure 4.18 Class Diagram.....	56
Figure 4.19 TruAwake Sequence	57
Figure 4.20 TruAwake Activity Model.....	58
Figure 4.21 State Model of TruAwake.....	59
Figure 4.22 Component Diagram for TruAwake Application... ..	60
Figure 4.23 Physical View: Deployment Diagram	61
Figure 4.24 TruAwake ER Model.....	62
Figure 6.1 App Testing.....	75

List of Tables

Table 2.1 Physiological Signals	11
Table 2.2 Possibility of Combining Various Measures.....	14
Table 2.3 Comparison Between Existing Studies	15
Table 2.4 Comparison of Existing Studies and Proposed System	17
Table 3.1 Use Cases with their Description... ..	26
Table 3.2 Estimated Funding Table	34
Table 3.3 Human Efforts and Contribution in TruAwake	35
Table 4.1 Technology Stack for TruAwake	43
Table 6.1 Test-Case 1: Camera Testing	68
Table 6.2 Test-Case 2: Facial Land-mark	70
Table 6.3 Test Case 3: Drowsiness Detection Accuracy.....	71
Table 6.4 Test-Case 4: Alert Mechanism	72
Table 6.5 Test-Case 5: History Recording	73
Table 6.6 Test-Case 6: Search Features	74

CHAPTER 1
INTRODUCTION

1.1 Introduction

The TruAwake System desire to accommodate the platform that provides a real-time safety system. This platform is designed to monitor drivers and prevent accidents caused by drowsiness and fatigue. Through AI power, machine learning, computer vision and sensors, this system detect signs of drowsiness and generates alerts to keep the driver safe and awake.

This platform is life-saving system that monitors the driver's facial expressions like yawning, eye closure, head tilts, body movements and alert the driver by triggering alarm, and break reminders. Different activities also included in application which keeps the driver alert.

Additionally, System also includes the GPS which monitors the speed and allow driver to set location, by ensuring a more safer and responsible driving experience. However, on motorway and on long highways, TruAwake ensures that you stay safe, by reducing the drowsiness and fatigue related accidents Stay Alert, Stay Alive with TruAwake(Driver Drowsiness Detection).

1.1.1 Problem Domain

Nowadays, drivers' drowsiness and fatigue affect road safety, and hundreds of lives are lost every day due to drowsiness and fatigue-related accidents, especially on motorways and highways in Pakistan. To enhance road safety and save human lives, we need to develop a system that alerts the driver through real-time monitoring [1].

1.1.2 Proposed System

Our proposed system, TruAwake, is mobile application designed to detect driver drowsiness and fatigue. It uses smartphone cameras and built-in sensors with AI-powered image processing to monitor facial expressions, especially eye closure, yawning, head tilts and body movements helping to identify signs of drowsiness and fatigue. By providing real-time alerts to drivers, the app helps prevent accidents and improve rad safety. The mainly focus is on making the solution efficient, accessible, and user-friendly.

1.2 Motivation

The increasing numbers of road accidents caused by drowsiness and fatigue in Pakistan especially on motorways and highways highlights the urgent need for safety improvement solutions. Promoting awareness about the issue is important to inspire safer driving habits.

By utilizing technologies such as AI powered image processing and smartphones built-in sensors we can create smart solutions like TruAwake application helps to detect early signs of drowsiness

and monitor alertness in real-time. Regular updates and improvements will make the software even more effective and efficient by adding new features and keeping up with new technologies.

1.3 Problem Statement

Driver drowsiness and fatigue are major causes of road accidents, resulting in severe injuries, with many lives lost instantly. There are many traditional ways to detect driver drowsiness and fatigue, such as hardware sensors, intrusive monitoring systems, wearable devices, and built-in vehicles like those in Tesla.

However, these methods are often expensive, impractical and not easily accessible for many drivers. There is a need for developing affordable and non-intrusive solutions that can detect drowsiness using easily accessible technologies.

1.4 Aims and Objectives

- Detecting driver's drowsiness and fatigue
- Provides real-time alerts
- Ensures accessibility and affordability
- Enhance road safety
- Uses AI and Built-in smartphone camera sensors
- Improves user experience
- Give user-friendly platform

1.5 Scope of the Study

TruAwake, an application designed to improve road safety and reduce the cost of hardware. It is a mobile based application which uses smartphones camera and built-in sensors to monitor drivers drowsiness and fatigue and generates real-time alerts to the drivers.

Utilizing AI-powered image processing technology, it analyzes facial expressions like eye closure, yawning, head tilts and body gestures, the key signs of drowsiness. It continuously monitors drivers' facial expressions and generates alerts when drowsiness is detected.

Alarm triggered for fewer seconds and if the driver doesn't get alert the application informs the motorway helpline therefore, they would be able to reach on time before meeting any mishap. This application uses artificial intelligence to enhance accuracy and to reduce false alerts. It is cost effective and non-intrusive solution for the problem, it is designed for ease of integration, accessible and affordable for personal cars, commercial fleets and public transport as well. By

combining AI image processing and smartphones built-in sensors, this application offers a simple yet effective way of reducing drowsiness and fatigue related accidents.

1.6 Process Model

TruAwake, an application that follows a structured process model to enhance road safety and to ensure efficient drowsiness detection and alertness. The model process of TruAwake begins with:

1.6.1 User registration and Authentication

To get access to the dashboard driver or fleet owner must create an account by adding their personal details and then gets verification email or SMS.

1.6.2 Data collection

Once user successfully accesses the dashboard, utilize smartphones camera and built-in sensors to monitor facial expressions such as eye closure, head movements, yawning and as well as body movements.

1.6.3 AI-driven Analysis

Once the data is collected it can be analyzed through AI-driven analysis, using machine learning models to assess the signs of drowsiness by identifying abnormal expressions and patterns.

1.6.4 Alert System

Once the drowsiness is detected, an alert system triggered sending immediate warnings to the driver through alarm sounds and it also sends alert messages to driver for making them alert.

1.6.5 Post-driving report

The analyzed data of detected drowsiness incidents and driving behaviors can be stored in the app and allows the user or fleet owner review reports of their driving behaviors or patterns.

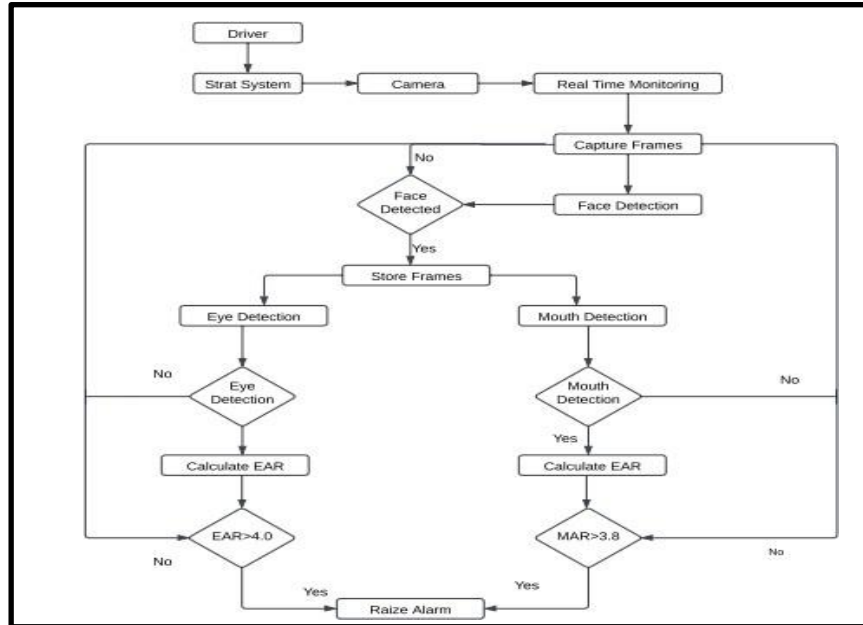


Figure 1.1 Working Flow of TruAwake

1.7 Proposed System Features

Our proposed system (driver drowsiness detection) is mobile based application system which contain different features. It features real-time driver drowsiness detection using smartphones camera and built in sensors to monitor facial expressions and driving patterns. AI-powered image processing detects early signs of fatigue and drowsiness and triggered the alert system.

Additionally, app also provides the post driving reports, allowing users to review their driving patterns. All these features makes the TruAwake smart, cost-effective and non-intrusive solution to improve road safety.

1.8 Development Methodology

The development methodology of TruAwake uses Agile model ensuring feasibility and continuous development. The development process starts with requirement gathering or analysis where user sets and analyze the project goals. Then the second phase is design, where user makes UI/UX design of an application which helps to work according to the finalized design. In development process application goes through many phases, where each iteration undergoes the testing and evaluation process.

TruAwake(driver drowsiness detection) system with graceful and repetitive development phase followed by continuous monitoring and updates to improve detection accuracy and user experience. Regular updates in application are based on users feedback and new technologies.

1.9 Implementation

1.9.1 Technology Stack

Our frontend is developed using React native as well as Visual Studio code and android studio for debugging and emulation. Image processing AI modules are being implemented by means of Google API's and JavaScript based backend services.

1.9.2 Frontend Development

For designing an intuitive UI/UX of the system, we consider the React Native that is built on the premise of JavaScript and TypeScript to create mobile-based applications. There are components like view, text, image and stylesheet which are utilized to make mobile screens to be responsive.

1.9.3 Backend Development

We integrate AI-driven drowsiness detection using smartphone cameras and built-in sensors. Using mobile sensors data will proceed for real-time monitoring. The processed data will be stored in secure database for the reviews.

1.9.4 Testing & Debugging

We conduct unit testing for individual components for better outputs. Using different driving conditions, we perform real-time testing for better driving results. By optimizing AI modules, reducing the false results.

1.9.5 Deployment & Maintenance

We release our application on Google Play Store and provide regular updates based on users feedback and with the change of technology to enhance the user experience.

1.10 Nature of Project

TruAwake, an application designed to prevent drowsy and fatigue related accidents utilizing AI-driven image processing. The system uses smartphone cameras and built-in sensors to detect earliest signs of drowsiness(yawning, eye closure, head tilts) and generates real-time alert to the driver.

- **Domain:** The domain of TruAwake application is AI-based road safety and monitoring driving patterns.
- **Technologies Used:** Application uses mobile sensors (camera, Accelerometer), and computer vision to detect drowsiness and fatigue and to enhance road safety.
- **Platform:** Using React Native for frontend, Google API's for AI modules and JavaScript for backend we develop the TruAwake(Driver Drowsiness Detection Application).

It is a non-intrusive, cost-effective and mobile-based solution for the addressed problem of existing high cost, intrusive and hardware which are not accessible. This application can be further integrated into personal and commercial vehicles, as well as fleet managements.

1.11 Overview

TruAwake, an AI-powered driver drowsiness detection system designed to enhance road safety by avoiding drowsiness and fatigue related accidents specially on ,motorways and highways in Pakistan. By leveraging computer vision, Google API's and smartphones sensors, it monitors facial expressions(such as eye closure, head tilts, yawning, and body movements) to detect early signs of drowsiness and fatigue. Application provides instant alarming, break reminders, alert notification, activities and many more to keep driver awake, active and responsive. Different from existing hardware, expensive and intrusive solutions, TruAwake provides a cost-effective, non-intrusive and affordable, mobile-based solution that is easily accessible to all drivers. Additionally, system integrates the GPS monitoring system to track speed and location as well generating post-driven reports for drivers to review their driving patterns and their behavior.

Application will developed by using React Native, Google API's, JavaScript and follows an Agile methodology to ensure the continuous improvements based on user feedback and change of technologies. TruAwake provides an efficient and user-friendly platform to reduce the drowsiness-related accidents and make the roads safer for everyone.

CHAPTER 2
EXISTING SYSTEMS/ RELATED WORK

2.1 Introduction

Road accidents prove to be the most substantial hurdle in transportation. A World Health Organization (WHO) release shows that approximately 1.19 million people die every year in road accident and 20 to 50 million people suffer non-fatal injuries, with many incurring disabilities [2].

The statistics from Road accidents prevention represents a fundamental reason behind researchers dedicating efforts into driver drowsiness detection system during recent years in Pakistan as well as other countries like USA. etc. multiple approaches exist for resolving this matter which illustrate varying capabilities in problem resolution. The currently available solutions for drowsiness detection categorize themselves into physiological monitoring system, behavioral analysis as well as vehicle-based detection methods [3]. Each solution shows specific challenges when used for adjusting to different driving scenarios. This documentation aims to propose a system for driver drowsiness using AI-driven image processing and smartphones sensors that will help to detect a drowsy state at an early stage.

2.1.1 Background

During drowsiness a person progresses from alert state to sleep while showing impaired concentration and an inability to keep their eyes open and unstable body movements and frequent sternocleidomastoid muscle contractions known as yawning. The inability to respond rapidly and handle a vehicle becomes severely compromised in people with this condition which leads to higher peril of traffic accidents.

People generally experience drowsiness because of circadian rhythm when they drive at night between 12:00 AM and 7:00 AM and in the afternoon between 2:00PM and 4:00PM. The risk of drowsiness accidents increases among solo young drivers from the age bracket of 18 to 30 on motorways and highways. The detection of sleepiness during driving proves difficult mainly because drowsiness remains undetected, so it appears alongside mechanical equipment breakdowns and traffic law infractions [4].

What is essential for this situation is a detection system that functions at an early stage and provides affordable and non-intrusive solutions. Present drowsiness detection solutions exist but they are expensive and intrusive and fail to provide security.

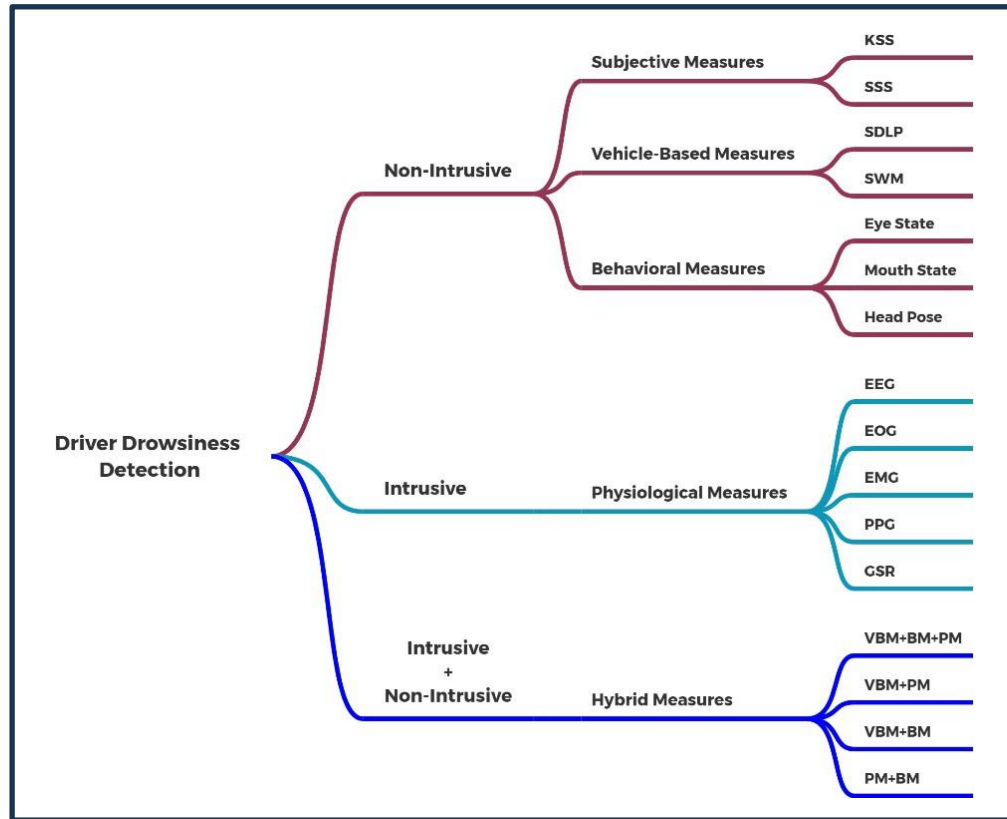


Figure 2.1 Driver Drowsiness Measures

Source [5]

2.2 Types of Existing Systems

Several types of Driver Drowsiness Detection System Exists; each solution is different from other some of them are given below:

2.2.1 Physiological Monitoring System

Driver drowsiness detection relies on physiological monitoring systems because these systems directly measure changes in physical and mental state of the driver. Vehicle-based data and behavioral cues are absent in physiological methods which instead study biological signals that change during fatigue development. These warning systems detect drowsiness during its early stages when drivers demonstrate no visible signs therefore giving enough time to prevent accidents. Ultimately, they protect both drivers and others on the road.

Physiological monitoring system incorporates wearable sensors which detect drowsiness-related psychological signs through EEG headbands as well as heart rate monitors. High accuracy characterizes these systems, yet they remain costly along with being relatively unpleasant for continuous use [6].

2.2.1.1 Types of Physiological Signals

Table 2.1 Physiological Signals

Physiological Signals	Description	Drowsiness Indicator
Electroencephalogram (EEG)	Used to detect brain wave activity makes it an extensively recognized method.	Increase appearance of alpha and theta which act as decreasing alertness in individuals.
Electrocardiogram (ECG)	Monitors heart rate and heart rate variability (HRV).	Decreases in heart rate and slower HRV.
Electromyogram (EMG)	Detect muscle activities, head tilts and facial actions.	Decreases muscle tension signifies ongoing drowsiness and fatigue.

2.2.1.2 Advantages of Physiological Monitoring

- Apart from external symptoms the system detects fatigue early through internal physiological monitoring.
- The system functions effectively across different driving conditions without any impact from lighting weather or road type because it differs from vision-based systems.
- The system bases its drowsiness evaluation on actual measurements of physiological processes throughout brain and heart and muscles.
- Modern wearable technology enables researchers to include sensors within friendly devices such as smartwatches or headbands.
- The system protects against accidents through its ability to notify drivers about drowsiness and fatigue moments.

2.2.1.3 Disadvantages of Physiological Monitoring

- Long-term sensor use becomes uncomfortable when they require driver contact through skin and scalp surfaces.
- High-level filtering systems need implementation to process unwanted signals stemming from vehicle motion or road vibrations together with environmental disturbances such as perspiration.

- The implementation of EEG, EMG, ECG systems present a challenge because such equipment comes with high price tags and scalability limitations.
- Technical specialists with specialized expertise must handle the installation of signals and their continuous maintenance and signal interpretation processes.

2.2.2 Behavioral Analysis Systems

Behavioral detection systems track driver sleepiness through direct observation of face-related activities which include eye movement along with mouth activities and movements of the head. Eye blink rate and the percentage of Eye Closure (PERCLOS) operate as a principal sign through ML and DL algorithms. These non-invasive detection techniques find regular users during simulated and actual driving operations.

Driver's face features become apparent through cameras which usually attach to the dashboard. Each phase within detection has three primary steps starting from data acquisition through feature extraction before classification. The system acquires video and image inputs then performs facial feature analysis through preprocessing to classify drivers as either alert or drowsy.

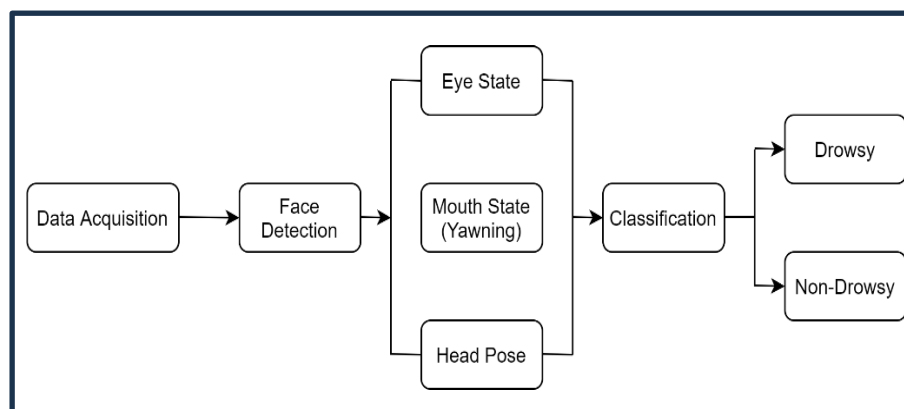


Figure 2.2 Driver Drowsiness Detection Using Behavioral Measures
Source [5]

2.2.2.1 Advantages

- The system allows drivers to maintain a comfortable experience while using the system due to its non-touch functionality.
- Driver behavior features enable real-time analysis which happens during continuous operation while driving remains uninterrupted.

- Uses affordable hardware such as standard cameras.
- Suitable for both simulated and real-world environments.

2.2.2.2 Disadvantages

- Video-based operate poorly within conditions of insufficient illumination or dark periods throughout the day and night.
- The detection of facial features is hindered when drivers wear glasses or masks together with facial hair or any type of headwear.
- Behavioral measures alone are insufficient because they lack reliability in some circumstances so detection systems integrate them with vehicle-based or physiological methods.

2.2.3 Vehicle-Based Detection Systems

Vehicle-based detection systems used to analyze driving patterns including steering patterns and braking behavior together with lane deviation for identification purposes. The technology works effectively but needs built-in in vehicle equipment which become impractical for certain devices.

Current driver drowsiness detection systems in vehicles function by executing persistent unnoticeable monitoring of driver performance and alertness through combining driving behavior signals and physiological signs. The primary function of these systems depends on tracking driving performance indicators which include measurements of steering wheel movement and lane position drift and lateral acceleration signals for spotting initial signs of drowsiness.

Additional psychophysiological measures that include eye closure rates become integrated in certain systems [7]. The system detect tangible performance indicators which indicates drowsiness onset before sleeping emerges allowing drivers to receive prompt alerts.

The detection process typically involves:

- The system gathers information from vehicle monitoring hardware which includes steering angle and lane tracking abilities.
- The extraction of features occurs through the analysis of driving patterns.

- The system classifies drowsiness levels through automated algorithms that have been trained for this purpose.
- The system triggers driver warnings by means of visual and auditory components as well as tactile prompts to prevent crashes.

2.2.3.1 Advantages

- The system performs measurements without interrupting drivers in their routine activities.
- The system tracks the driver performance along with vehicle conditions permanently throughout operational periods.
- The system maintains effectiveness while not touching the driver to ensure monitoring operations.
- The system proves compatible with other systems that aim to reduce accidents such as infrastructure-based alerts and distance-based alert systems.

2.2.3.2 Disadvantages

- The system's performance operates differently according to the quality of lane markings as well as weather conditions and heavy traffic levels.
- Initializing cost together with system complexity leads to expensive expenses.
- The system primarily functions effectively above 50 mph speed but only works properly in rural areas and highways.
- Existing technological systems assist in road safety enhancement but face problems because of their expense and requirement of hardware, intrusive, costly and privacy-invading elements.

Table 2.2 Possibility of Combining Various Measures

Hybrid Measures	Accuracy	Advantage	limitations
Vehicle-based +Behavioral	91%	Easy to use	High false detection and dependent on geographical conditions

Behavioral + Physiological	98%	High accuracy	Highly intrusive
Vehicle-based + Physiological	93%	High accuracy	Highly intrusive
Physiological + Behavioral	81%	High accuracy and ease to use	Expensive and more challenging to implement

Source: [5]

2.3 Existing Systems/Related Work

This table is summarizing the features, strength and limitations of exiting work. It presents a comparative analysis of existing work and our proposed solution that utilize technologies like facial recognition, OpenID to prevent cyber-attacks.

Table 2.3 Comparison Between Existing Studies

Features	Real-time driver drowsiness detection (2019) [12]	Smart Eye Pro7(2020) [13]	Hybrid Driver Drowsiness Detection(2021) [14]	EEG and Head Movement Detection(2023) [15]
Detection Method	Facial features(camera-based)	Eye/yawn tracking	Image and biological signals	EEG head band and accelerometer
Technology	Computer vision(OpenCV, Dlib)	Proprietary Hardware sensors	Multimodal machine learning fusion	Brain signals And motion tracking
Strengths	High early detection accuracy	Industry-grade robustness	Very high accuracy	Highly accurate with physiological signals
Limitations	Depends heavily on camera quality and lightning	Expensive hardware installation needed	Complex, Multiple data Sources and sensors required	Headband discomfort, expensive setup

Hardware Requirements	Good quality camera	Dedicated Sensors and installation	Sensors for biological signals	EEG head band and accelerometer
Cost	Depends on camera	Very hard hardware cost)	High	High cost of EEG devices
Data Processing	In-device or server	Server/cloud-based	Server/cloud-based	Server / cloud-based
Personalization	No	No	No	No
Environmental Adaptability	Sensitive to lighting	Works in various lighting but costly	N/A	N/A

Source: [16]



Figure 2.3 Smart Eye Pro and EEG signals with dynamical encoder and decoder

2.4 Comparison of Past Papers with our Project Idea

Here is a structured table based on the research papers, evaluating them across key features:

Table 2.4 Comparison of Existing Studies and Proposed System

Research Paper/ Systems	DM	CE	SSL
Real-time driver drowsiness detection(2019) [12]	Eye &facial features	Medium	High
Smart Eye Pro7(2020) [13]	Eye, yawn tracking	X	High
Hybrid Driver Drowsiness Detection(2021) [14]	Multimodal fusion	X	Very high
EEG and Head Movement Detection(2023) [15]	EEG headband and accelerometer	X	Very high
Enhanced CNN- Based Detection(2023) [17]	Deep learning models using CNN	Medium	Medium

2.5 Explanation of Important Constructs of the Application

2.5.1 Domain

The TruAwake application functions in intelligence driver assistance systems territory to detect drowsiness in real time through mobile-based artificial intelligence (AI) technologies. The entire system operates accurately through various key constructs which provide non-intrusive and cost-effective functionality.

2.5.2 Drowsiness Detection

The central concepts of this application serves as its main operational purpose. The smartphone's built-in camera detects drowsiness through facial analysis of such signs as eyelid closure, head tilts, body movements, and yawning.

Driver alertness classification operates through machine learning algorithms that specially use Google API's to process the system inputs in real time.

2.5.3 Mobile Sensor Integration

The application exploits smartphone hardware including camera, acceleration, and gyro sensors via react native sensor libraries to record the real-time movement of the phone and facial expressions.

The detection system acquires data about head position alongside movement records along with ambient lighting information which strengthens its capacity for detecting fatigue before it becomes critical by eliminating the need for any additional hardware.

2.5.4 Image Processing

Facial recognition together with tracking technology processes the inputs from cameras. By employing Google API's and facial landmark detection of eyes along with eyebrows and lip positions allows identification of driver drowsiness and fatigue signs.

2.5.5. Real-Time Alert Mechanism

When the app detects drowsiness, it automatically activates notifications which send alerts via sound signals and popup messages with vibration and emergency calls. Fast driver reactions depend on the prompt feedback loop which serves as an essential measure to prevent possible car accidents. If driver does not react to the prompt feedback it can call to the motorway helpline.

2.5.6 Adaptive Learning

The system completes a module that teaches users about their driving patterns as time advances. TruAwake builds individual user profiles to establish their specific driving behaviors and drowsiness and fatigue characteristics which than leads to better detection of genuine drowsiness situations.

2.5.7 Location-Based Sensitivity

The application makes its detection threshold flexible through GPS tracking to raise sensitivity on highways compared to urban zones because highway accidents provoked by drowsiness are more prevalent.

2.5.8 Post-Driving Reporting

The application provides post-drive analysis which creates reports to display observed drowsiness events and documentation about their actions. The analytical data generated reveals insight that benefit both drivers who track their performance growth and fleet managers together with insurance partners who need evidence for better choices.

2.5.9 Privacy and On-Device Processing

User privacy remains protected because calculations run exclusively on the internal storage of the device users carry. The protection of user confidentiality remains intact through local processing since proprietary data remains within the device boundaries.

2.6 Summary

Millions of people lose their lives together with millions getting injured annually due to driver drowsiness which acts as a primary cause of road accidents throughout the world. Current detection systems for drowsiness divide into three categories which include physiological monitoring, behavioral analysis, and vehicle-based methods. Internal bodily signals provide high accuracy for physiological systems, but these systems come with both high cost and invasive procedures. The method of behavioral analysis that monitors eye movements along with yawning behavior utilizes affordable and comfortable solutions yet encounters problems from poor lighting and obstructed facial views.

Vehicle-based systems that monitor steering patterns together with lane deviations remain expensive and perform poorly under adverse weather conditions or heavy traffic. The detection capabilities of Smart Eye Pro7 and EEG headbands remain accurate, yet these systems face challenges because they require high setup costs together with discomfort to users during usage. Proposed system TruAwake addresses diverse issues through smartphone-front cameras that uses built-in sensors together with AI to offer real-time monitoring along with affordability and privacy protection. The system issues warnings about driver fatigue and drowsiness discoveries

while acclimating itself to individual user behavior patterns through time and creating driving reports which enhances its availability beyond present techniques.

CHAPTER 3

REQUIREMENT SPECIFICATIONS

3.1 Introduction

The development of TruAwake requires Requirement Specification as the initial phase to establish essential functional requirements alongside essential non-functional needs. The mobile driving safety system project requires precise requirement definition to achieve both performance targets and user expectations because the project targets a mobile driver safety system that costs less.

Throughout this phase developers receive specific executable work responsibilities from the proposed solution while valuable stakeholders maintain mutual comprehension of the process. The requirements include all the necessary hardware together with software needs and user interface procedures and data processing specifications as well as performance measurement criteria. TruAwake develops complete requirement specifications to produce a dependable system that operates in real time while protecting privacy and increasing roadway security by eliminating the need for costly specialized hardware.

3.2 Interface Requirements

System components need essential elements for effective communication with external environments and internal elements through interface requirements. It is essential to precisely define interfaces throughout the TruAwake application because it incorporates hardware together with software features using mobile platforms and smartphones.

The interface requirements section uses hardware and software requirements subsections to manage the system needs effectively. The most important part of this section is the design of the system that can support the functional requirements of the system and meet with the user expectations.

3.2.1 Hardware Interface Requirements

The functioning of TruAwake application depends on mobile hardware components within the device [20]. The hardware interface requirements are:

3.2.1.1 Smartphone Camera

A front-facing camera serves as essential hardware for capturing current driver's face images for the system. SD cards or cameras used by the system

application need to feature HD (720) resolution to detect eyes along with mouth and head movement's reliability.

3.2.1.2 Accelerometer and Gyroscope Sensors

Mobile sensors must detect three indicators of drowsiness by monitoring head tilts and posture movements together with micro-movements of the person.

3.2.1.3 Minimum Device Specifications

Android smartphone or tablet with at least 2GB RAM. The system requires processors with a minimum quad-core specifications to execute AI processing operations.

3.2.1.4 Storage & Operating System

- Minimum 300MB available space for app installation and data storage.
- Android 5.0(Lollipop) or higher.

3.2.1.5 Device Mount or Holder

The camera system requires an installation device or mounting system to hold its position to prevent manual operation when driving.

3.2.2 Software Interface Requirements

Through its software interface different software modules integrate perfectly with third-party libraries and operating system functions. The main requirements for software interface components include:

3.2.2.1 Operating System Compatibility

The application needs to be compatible with Android OS version 5.0(Lollipop) through the latest release.

3.2.2.2 Development Environment

The application can be developed with React Native with the options of android studio and visual studio code. It is developed with JavaScript and TypeScript mobile development with native android modules as needed. JavaScript or other mobile inference engines are used for the implementation of AI modules.

3.2.2.3 Cloud or Backend Services

The authentication system uses Supabase Authentication to provide secure user login management functions.

3.2.2.4 API Integration

The Location-Based Analysis capability of Google Mapas API includes an adjustable sensitivity level which depends on the driving location.

3.3 Functional Requirements

In this section we define the key functionalities of the recommended system for drivers.

3.3.1 User Authentication

Users(drivers) must access an easy and secure signup process and authentication system through TruAwake application. Users need to enter essential information for creating their verified account while they through the registration process.

3.3.2 Real-Time Facial Monitoring

The application needs smartphone front camera technology to observe driver facial features including eyes, mouth and the movements of their head in real time. This system performs live detection of driver fatigue indicators including eye blink duration and yawning events together.

3.3.3 Drowsiness Detection Algorithm

The application depends on AI-based Google API's to examine real- time facial images as a processing method. Timely Alerts are triggered when the system identifies the signs of drowsiness through the combination of prolonged eye closure with yawning and head drooping.

3.3.4 Sensor Data Collection

The app gathers movement information through smartphone sensors which include both the accelerometer and gyroscope. Posture changes along with head movements flag occurrences of sleepiness and lowered attention to the system.

3.3.5 Alert System

The program generates instant alerts by visual and audio or vibration methods when it detects drowsiness. The application gives serious warning notices to drivers to pull over and rest when severe fatigue threatens accident prevention.

3.3.6 Location-Based Alert Sensitivity

GPS data enables the app to monitor and adapt detection sensitivity according to what kind of environment the driver operates in. The sensitivity of the app will adapt to reduce drowsiness detection on roads which experience higher rates of drowsiness-related accidents.

3.3.7 Post Driving Report

During every commute the application generates a written summary containing instances of drowsiness detections together with recorded behavior patterns and the number of alerts issued. Through the feedback system people who drive can better identify risky behaviors for better safety.

3.3.8 Data Privacy

The device will handle all local storage and processing activities concerning personal data such as sensor readings and facial images. The system stores data locally on the device yet it sends data remotely to external locations only after user permission is granted.

3.4 Use Case Model Process

3.4.1 Actors:

3.4.1.1 Driver

The main user of TruAwake application is the driver. The application requires users to log in before granting permission for start monitoring the driving session. The application together with head movement patterns in current time.

While running in the background mode the driver does not have to interact with the app as it automatically monitors the driving scenario. The app activates warning signals by way of alarms and vibrations when it detects sleepiness indications through yawning combined with eye closure duration or odd head movements.

When signs such as frequent yawning, extended eye closure, or unusual head tilts are identified, the app immediately triggers warning signals. These include loud audio alarms and device vibrations to alert the driver and prevent potential accidents.

3.4.1.2 Systems (App)

All intelligent features of TruAwake rely on the system(app) which operates within the application. The System uses ML models to extract data from the Smartphones camera and sensors then evaluates the processed information with AI algorithms for detecting signs of drowsiness and fatigue. All detection models executed directly on the device maintain both privacy standards and run at maximum speed.

The data collection system sends information to artificial intelligence(AI) algorithms which assess driver attention level. AI algorithms receive training to discover small signs indicative of driver drowsiness various behaviors such as continuous yawning combined with prolonged eyelid closure and brief sleeping events. User devices process the detection models directly since the algorithms do not require cloud-based operation. The method runs directly on users devices to provide instant processing together with enhanced security by maintaining all data within the device storage.

3.4.1.3 Admin/Owner

Under the application the optional Admin/Owner manages user accounts that exist within the system. The Admin/Owner role brings value to commercial fleets because such organizations use TruAwake with multiple drivers under their control management.

Through their role the Admin has capabilities to add new drivers to the system while tracking their drowsiness records and assessing overall device use. System administrators can modify alert specification together with sensitivity levels for alerts and establish user authorization privileges.

Table 3.1 Use Cases with their Description

Use cases	Description
Login/Registration	Users can access services by signing up or utilizing their existing accounts

Capture Driver Data	The system tracks eye motions and head position together with posture through built-in camera
Analyze Behavior	This system utilizes AI technology to identify driver drowsiness through detecting eye closure alongside yawning activities
Generate Alert	The system triggers an alarm together with emergency notifications when recognizes drowsiness
Post-Driver Report	Users can access a summary about driving activities and fatigue symptoms from their trip
Location-Based Sensitivity Adjustment	The device adjusts its capabilities through GPS monitoring particularly on motorways and highways
Environmental Awareness	Operation behavior changes depending on the lighting conditions and weather conditions
View Reports	The system enables users to check all recorded alerts and reports through out their driving seasons

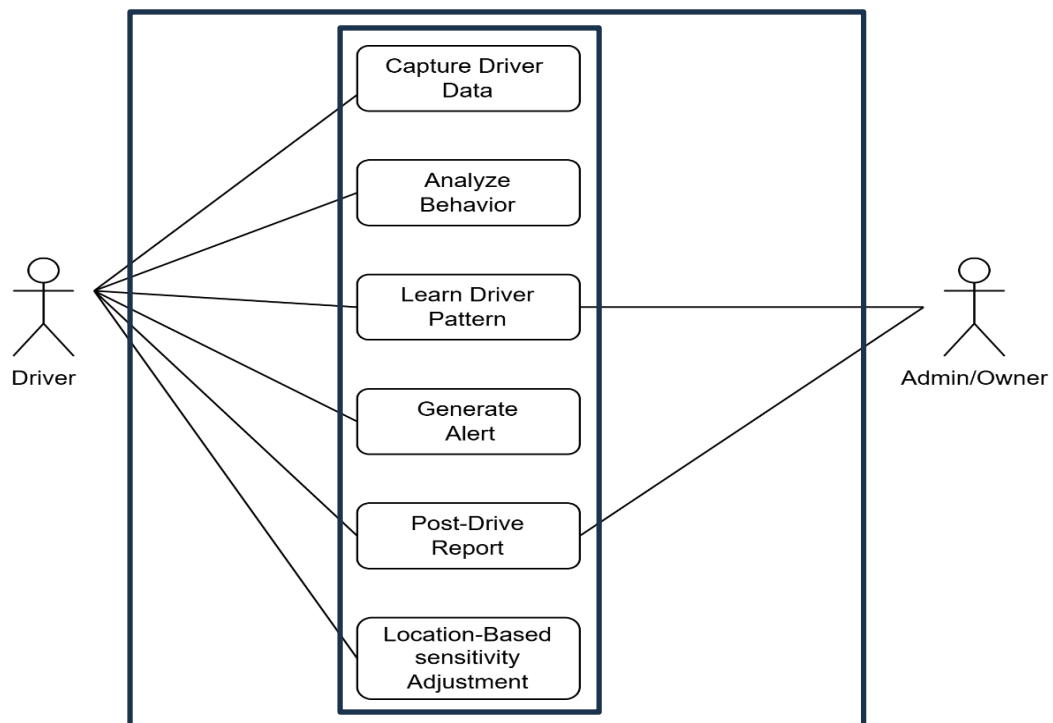


Figure 3.1 Use Case Model of TruAwake

3.5 Use Cases

3.5.1 Use Case Description

A Use Case Diagram is a graphical presentation that draw the connections between drivers (actors) and a software (the application or software), displaying the opposed process, working, and roles regulation within the software. For the TruAwake application constructed using React Native the Use Case Diagram represent the different working connections with the software and the relevant use cases or processing they can execute.

3.5.1.1 Use Case 1: User Login

Driver access to personalized drowsiness detection services runs through the user login use case after they open TruAwake app through login with their credients. Account registration needs to take place before the driver utilizes this process. The driver opens the TruAwake application which requires them to provide their username along with their password before proceeding.

The system validates the entered credentials through its password before process. The system validates the entered credentials through its authentication process. The system verifies the driver credentials successfully before granting entry to the application dashboard. System gives the two options after failed login attempt which include retrying the login process or initiating the password reset procedure.

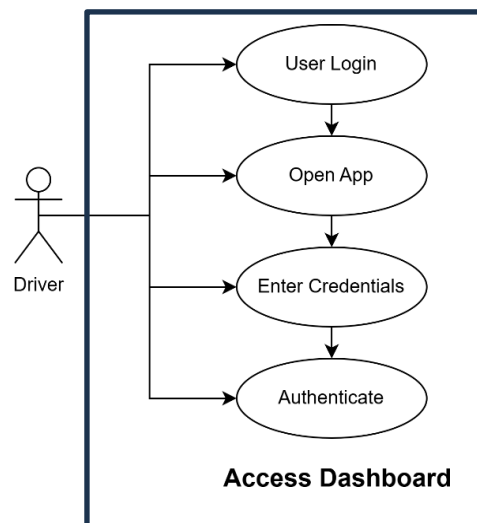


Figure 3.2 User Login: Access to Dashboard

3.5.1.2 Use Case 2: Drowsiness Detection

Through the Drowsiness Detection use case the Driver and System collaborates for monitoring driver fatigue indications. After activating the TruAwake app for active monitoring the system powers up the device camera and necessary sensors to begin.

The system maintains uninterrupted observation of all driver movements involving their eyes alongside face and head positions. The application will generate an instant alert through an alarm whenever the app identifies driver fatigue symptoms. When the system faces difficulties such as poor lighting conditions or blocked driver's facial view it can automatically modify settings including brightness level and sensitivity to enhance detection performance.

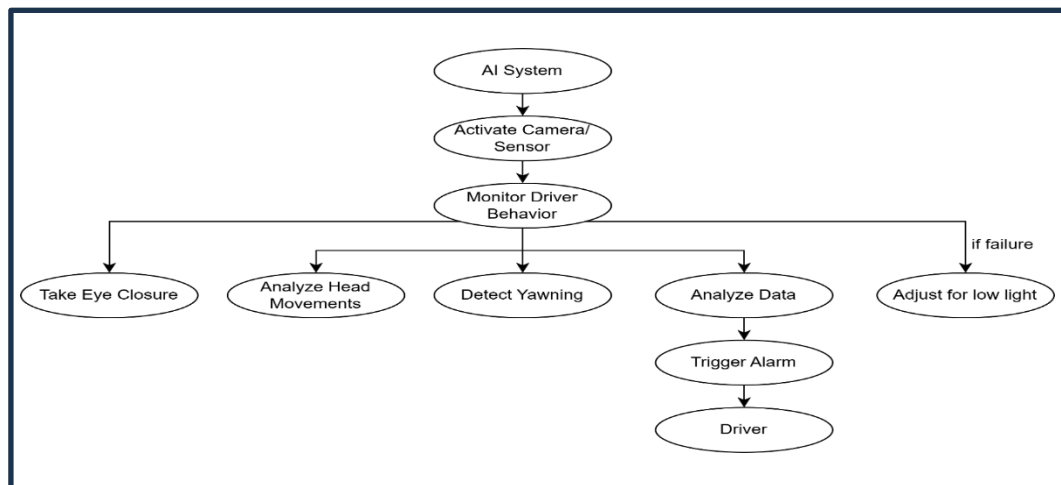


Figure 3.3 Driver Drowsiness Detection Pattern

3.5.1.3 Use Case 3: Alert Generation

The Generate Drowsiness Alert use case enables the system to take a forward-looking approach toward driver safety because it issues quick warnings after identifying drowsiness indicators. When the AI algorithm detects driver fatigue the system dispatches alerts via noise alerts and device vibrations and visual display signals to warn about the symptoms.

An emergency call function may get activated automatically based on the drowsiness severity level detected by the system. The driver receives alerts which they must react properly to maintain alertness either by stopping at a safe location

or taking a break or other necessary steps. Premier hazard mitigation starts when drivers reach a stage where they show signs of sleepiness thus preserving driving safety.

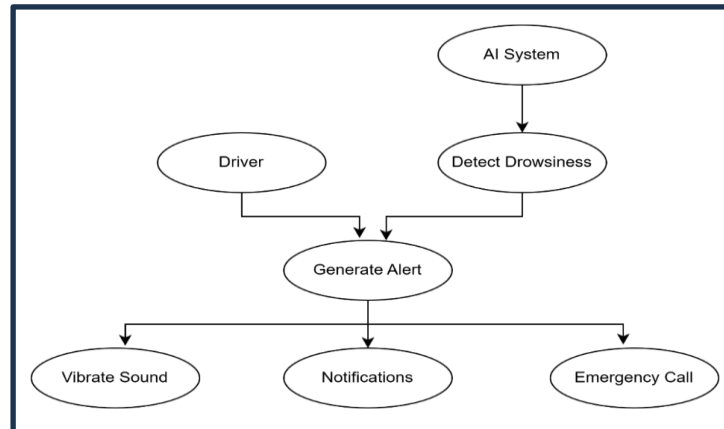


Figure 3.4 Alert Generation When Drowsiness Detected

3.5.1.4 Use Case 4: Post Driving Report

The Driver accesses their post-drive performance analysis including drowsiness-related information by using the View Post-Drive Report user case. The system produces a complete report after sessions conclude which contains details about drowsiness alert frequencies alongside driver conduct during driving time and their level of alertness throughout the journey.

Through review of this report the Driver becomes able to grasp when and how drowsiness developed to establish patterns which help them make educated choices about their driving approaches or modify application settings for successive sessions.

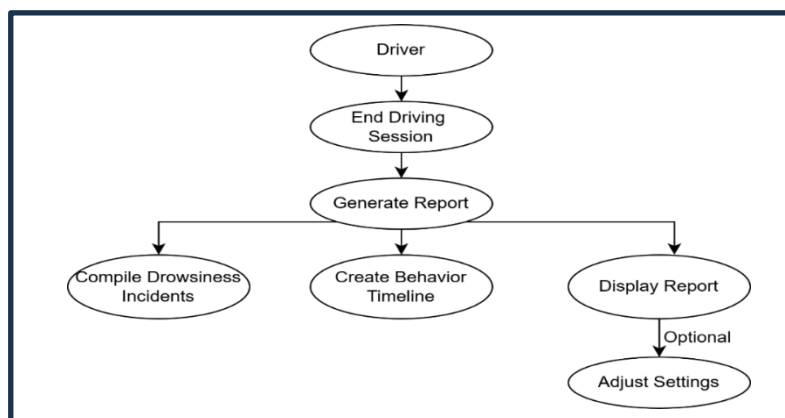


Figure 3.5 Post-Driving Report

3.5.1.5 Use Case 5: Manage Driver Account (Optional)

The Management Driver Account use case allows Admins or Owners to supervise all driver accounts present in the system. The administrator reaches the driver management area through the admin dashboard after login so they can create new driver profiles and modify existing account data or remove drivers from the system when necessary.

The driver record maintenance function allows the application to deliver precise and current data so users can receive personalized support, and monitoring can be more effective. When administrators finish these operations, the system automatically displays updated changes to allow proper registration or updates of drivers in TruAwake.

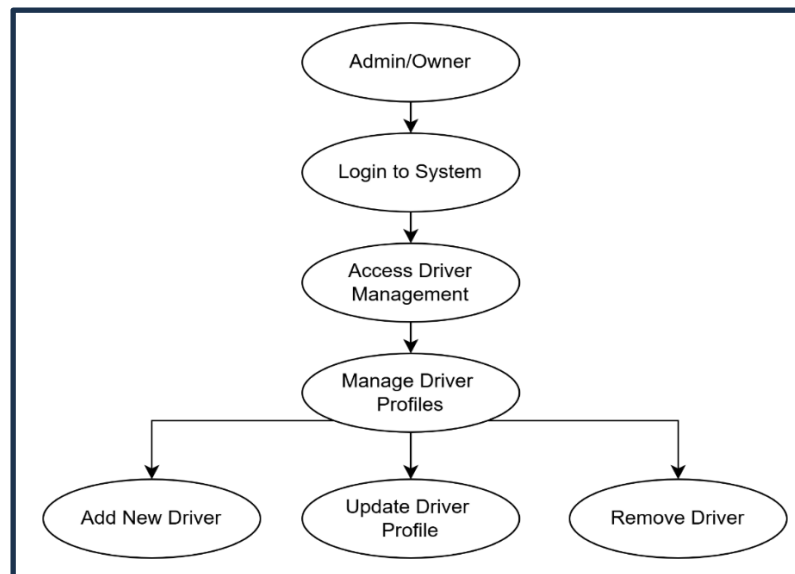


Figure 3.6 Driver Account Management (Optional)

3.6 Non-Functional Requirements

Systems performing under different circumstances are described by non-functional requirements that do not pertain to system functionalities. Non-functional requirements determine the quality standards together with usability, reliability and scalability of the TruAwake application.

3.6.1 Performance

The TruAwake application needs to work in real time mode. The TruAwake application should continuously monitor the smartphone's camera and sensors while running real-time processing without any detectable pause. The application needs to produce

warnings about drowsiness signs by presenting alerts in no longer than 1-2 seconds following detection of eye closure or yawning. The application needs to work without issues on devices that use at least 2GB RAM while staying power-efficient when users drive for extended periods.

3.6.2 Reliability

The system requires stable operation at the expected function levels. Regardless of slight adjustments in driver position or light availability the system needs to maintain its accuracy of operation. The application needs to retrieve monitoring data without data loss during phone calls or notifications or in cases where the application requires temporary closure. The system needs to identify normal driving patterns in order to prevent both problematic correct and incorrect detection instances.

3.6.3 Security

Security remains paramount when managing sensitive biometrics along video information. Every facial detection process together with analysis operates exclusively on the device to ensure user privacy remains protected. Secure login through password protection should be implemented in the platform. All stored or backed up data must be encrypted with accessible authorization restricted to a single user.

3.6.4 Consistency

TruAwake requires uniform operation in all android platforms which it supports. Users must easily identify familiar elements through standardized mobile interface patterns which they can understand. The application's alerts together with reports and detection results should display identical data output when used under identical circumstances. The device settings for sensitivity and thresholds should stay unchanged unless the user takes an explicit action to change them.

3.6.5 Usability

The system needs to provide an easy-to-use interface for users. During high-pressure situations or emergencies drivers need to understand the application without difficulty. The application needs to display alerts through audible sounds and vibrations and brief visual notifications. The driving reports must contain clear visual presentation that avoids complicated technical wording.

3.6.6 Maintainability

The maintenance and update capabilities of TruAwake should be simple to manage. Future improvements to TruAwake should follow a modular design which enables developers to add detection models and wearable sensors and redesign the user interface independently from the other system modules. The system requirements for updates should allow users to handle bug repairs and enhance system performance efficiently.

3.6.7 Portability

The app needs to operate on Android devices starting with version 5.0(Lollipop) and subsequent ones. The system needs to adjust its display when operating on devices with various screen dimensions and resolutions. The initial Android design includes features that permit a future transfer of Ios platforms or connection with wearable technology.

3.7 Requirement Resources

3.7.1 Equipment

3.7.1.1 Hardware Requirements

The smartphone functions as the main computing unit by operating the TruAwake application while managing digital processing tasks. Two critical pieces of hardware are required during TruAwake app development and deployment.

3.7.1.2 Android Smartphone or Tablet (minimum 5.0 Lollipop, 2GB (RAM))

The smartphone application functions on typical mobile devices. Both the smartphone front camera and built-in accelerometer and gyroscope sensors monitor driver facial movements.

3.7.1.3 Mounting holder

A dashboard or windshield-mounted phone holder provides drivers with both stability and a clear view of their face for monitoring.

3.7.1.4 IR Camera Module (Optional)

The next version of the system will include IR-based camera technology as part of future development to increase accuracy when detecting drivers during nighttime conditions.

3.7.2 Software Tools and Technologies

The system operates with free available tools that make it cost-effective for users. The application supports Android operating systems because they provide extensive worldwide adoption in Pakistan. Device-based processing runs to protect user privacy and independent operation from internet requirements.

- Developer and test applications for Android systems using Android Studio.
- The application constructed with javascript , react native as its programming languages.
- JavaScript library provides real-time facial detection of eye movements and facial movements such as yawning.
- The storage solution consists of Supabase for user reports that are option in the cloud or local system.

3.7.3 Funds

The combination of purchasing an inexpensive Android smartphone and utilizing Supabase in a free tier makes project maintenance costs reasonable. Open-source tools including Google APIs constitute most of the available libraries. Because of its minimal cost requirements, the TruAwake system presents an opportune solution for academic or pilot-scale development within Pakistani scientific environments. The developers created TruAwake as an inexpensive application which anyone can access.

Table 3.2 Estimated Funding Table

Items	Estimated Cost(PKR)
Android Smartphone	20,000(Start)
Internet Access	500-1000(Per Month)
Cloud Service	Free
Miscellaneous	2000

3.7.4 Human Efforts

A proficient team of few members will deliver the project in the specified academic period through modular task definitions that enable sustained progress. Three main team roles partition the required human resources to develop TruAwake.

Table 3.3 Human Efforts and Contribution in TruAwake

Team Member	Responsibility	Estimated Man- Months
Project Manager	Planning and monitoring process	1.5
Android Developer	App Development and UI implementation	3
AI Engineering	Using Google APIs for facial analysis	3
Computer Vision Developer	Face/Eye/Mouth detection with JavaScript Libraries	2
UI/UX Design	Designing user-friendly layouts	1
QA Tester	Testing app function and bug fixing	1.5
Report Writer	Preparing technical reports and documentations	1

3.7.5 Database Requirements

Here is a comprehensive overview of the database requirements for the TruAwake application (driver drowsiness detection):

3.7.5.1 Primary Option: Supabase Real-Time Database

Real-time database service offered by Supabase Suits applications that need instant synchronization of data across various devices through its cloud-based solution. Supabase Realtime Database provides secure scalable functionality that seamlessly integrates with Android development tools [21].

This implementation of Supabase database offers multiple benefits to the system:

- All detection alerts and driving behavior logs and user preferences update immediately within the system to show the same information across all devices.
- The database solution of Supabase adapts to grow from small operations to wide-scale needs suitable for individual and cooperate clients.

- User data protection features embedded into Supa base include built-in authentication systems and data access rules which secure information at all times.
- The platform offers cross-device support that enables administrators to observe multiple drivers simultaneously which benefits fleet management sector as well as logistics companies and safety enforcement agencies.

3.7.5.2 Optional Backup

Data storage predominantly relies on Supa base but the application will in-cooperate room or SQLite to create a local cache that stores data until network is restored.

3.8 Project Feasibility

3.8.1 Technical Feasibility

Technology feasibility exists for TruAwake application because of the following factors [22].

3.8.1.1 Utilization of Existing hardware

The application works on standard Android phones with their built-in cameras and sensors such as the gyroscope and accelerometer to function without additional hardware.

3.8.1.2 Cloud-Based Data Management

The application benefits from cloud-based data management that enables synchronization through Supabase Realtime Database to support remote access and scalability.

3.8.1.3 Open-Source Libraries

The implementation of JavaScript libraries with Google API's enables developers to create cost-efficient solutions with simple deployment possibilities.

3.8.1.4 Real-Time Processing

The application performs data processing instantly to send immediate drowsiness warnings to drivers when detecting their sleepiness.

3.8.2 Operational Feasibility

The TruAwake application operates without disruptions in the driving environment according to its operational design [23].

3.8.2.1 User Friendly Interface

The application presents an easy-to-use interface which allows drivers to operate it without need for special training.

3.8.2.2 Real-Time Monitoring

The system tracks driver facial features and behavior patterns across time so it alerts users about drowsiness as it happens.

3.8.2.3 Immediate Alerts

The system sends direct audio or visual warning signals when it detects driver drowsiness to bring attention to the driver immediately.

3.8.2.4 Data Logging

The system function as a data logger for capturing detection with driver responses to analyze patterns which optimize detection algorithms.

3.8.3 Legal and Ethical Feasibility

The TruAwake application adheres to legal and ethical standards [24]:

3.8.3.1 Compliance with Regulations

The application meets all data protection regulations because it ensures user data remains private and secure.

3.8.3.2 Use Consent

The application obtains user permission after displaying clear data collection terms which become mandatory to continue processing user information.

3.8.3.3 Privacy Preservation

Data privacy is maintained because the application stores all information within the device which decreases the opportunity for unauthorized parties to access data.

3.8.3.4 Non-Intrusive Monitoring

The monitoring system tracks driver actions without letting users become distracted or feel uncomfortable to prevent safety distraction.

3.9 Summary

Reliable development of an effective driver drowsiness detection application depends on complete requirement specifications which Chapter 3 defines thoroughly. The design specifies how users must register and log in and how the application operates by using smartphone camera real-time facial monitoring and AI-based drowsiness indicator detection combined with intelligent alert options and behavior learning systems and post-drive reporting process. Non-functional requirements analyzed in the chapter specify that the system needs real-time performance alongside reliability through accurate detection and security features with on-device processing that protects user data by encryption standards alongside a user-friendly interface for drivers who use various Android devices.

The design requirements consisting of a mid-range Android smartphone with minimum 2GB RAM and phone mount along with optional IR camera modules to detect at night. The complete software stack comprises three main frameworks: Android Studio, JavaScript, React Native programming languages and Google API's for AI computation and for image processing and Supabase for database remote management. By defining the responsibilities of developers, AI engineers, testers, and project managers during a prediction of 13- man-months effort. This system maintains cost-efficiency because it depends on affordable hardware together with open-source libraries for execution in both educational student work and industrial-scale deployment.

CHAPTER 4

SYSTEM MODELING

4.1 Introduction

TruAwake application system modeling creates an organized framework to represent its drowsiness detection system elements along with their operational processes and data movement patterns. The system design shows how real-time driver fatigue detection occurs through the operation of mobile sensor inputs with data processing along with machine learning models and alert systems coordinated with user interface operations.

Using modeling of these systems we can evaluate operational functionality while detecting system weaknesses and validating that every component fulfills app safety targets. The approach enables future development and support tasks to run smoothly while enhancing communication effectiveness between technology professionals and users.

4.2 System Design

The mobile application TruAwake uses smartphone cameras and sensors with AI algorithms to monitor driver fatigue by identifying various tiredness signs such as eye closures and yawning and head movements. Real-time warnings come directly from the system while it operates under accessibility parameters with low costs and ensures user privacy.

4.2.1 Client Side(Mobile App)

The mobile application TruAwake operates on Android devices starting from version 5.0 and above through JavaScript/TypeScript or React Native programming technology. The app incorporates lightweight JavaScript-based AI libraries which enable it to monitor drowsiness by analyzing the front camera and sensors on the same device.

The application performs all of its functions locally on devices making detection possible without any internet connection. A phone needs to have at least 300MB storage and 2GB of RAM to operate TruAwake effectively.

4.2.2 Hardware Architecture

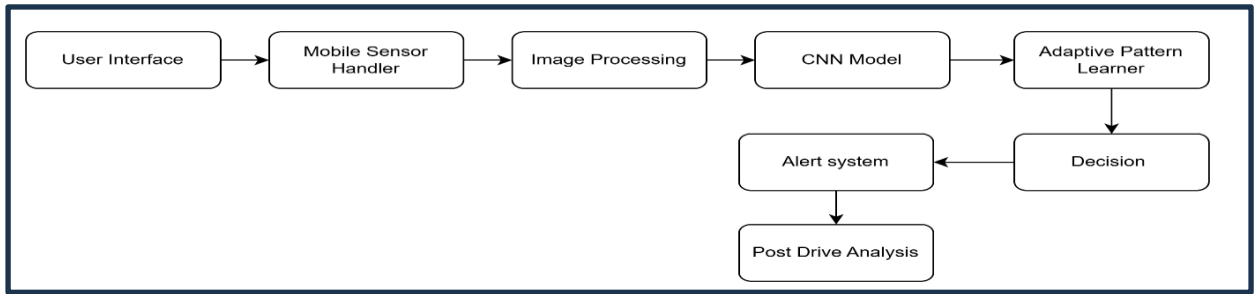


Figure 4.1 Hardware Architecture of TruAwake

4.2.3 System Modules

4.2.3.1 Login & User Management

The application enables safe user account creation and membership verification by using Supabase Authentication or JWT (JSON Web Token) to protect both passwords and data. The system implements a permission system that enables owners to perform separate tasks from Drivers within the platform. Users benefit from this configuration because it limits feature access to tasks that match their role thereby improving security and usability.

4.2.3.2 Camera & Sensor Interface

The system integrates the built-in camera to perform live facial recognition that tracks driver tiredness while driving. Through the accelerometer the device monitors abnormal head movements which serve as signs of driver fatigue. The system obtains contextual information about speed and location to improve accuracy of detecting drowsiness among drivers across different driving scenarios through GPS integration.

4.2.3.3 Drowsiness Detection Engine

The Drowsiness Detection Engine uses Google APIs to interpret three critical fatigue indicators that involve facial features such as blinking rate and eye openness and yawning movements. The normative motion detection system tracks driver behavioral changes through sensor inputs for drowsiness detection purposes.

4.2.3.4 Machine Learning and pattern recognition

By utilizing machine learning mechanisms, the system continuously learns driver habits which produces better detection results. Historical driving behavior patterns serve as models through this system which decreases false alarm frequency and differentiates dangerous care from standard driving practices.

4.2.3.5 Environmental Analysis

The system modifies its sensitivity according to evaluated environmental elements that include both day-night conditions and road classification together with weather data obtained through external Application Programming Interfaces (APIs).

4.2.3.6 Alert & Notification System

Real-time alerts sent by the system utilize alarms and device vibrations to warn drivers who are experiencing drowsiness at the moment. A critical situation that happens while driving will automatically initiate a phone call to an emergency contact previously registered by the user.

4.2.3.7 Post-Driving Report

At the end of each drive session the system produces an alert summary showing detected periods when drowsiness was detected thus allowing drivers to assess their alertness performance. The system creates visual displays to monitor behavior evolution throughout periods. The device serves all collected data within its local storage to protect user privacy and stop any unauthorized access.

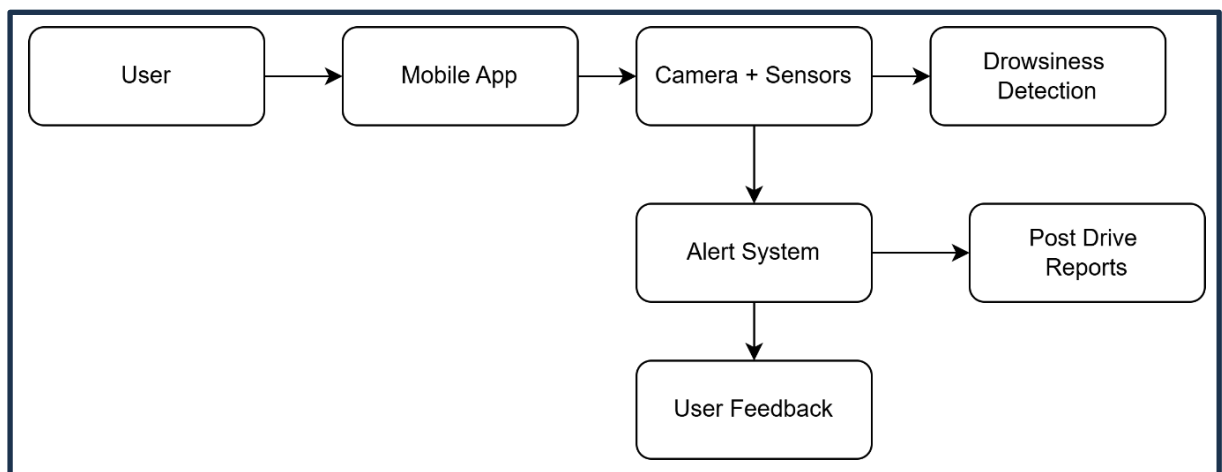


Figure 4.2 Data Flow Diagram (Level 0)

4.2.4 Limitations

- Inadequate lighting conditions together with eye covering devices such as glasses will impact on how precisely the app detects windiness.
- Driving conditions such as seat adjustment and brief periods of being awake result in irregular drowsiness detection performance from the system.

4.2.5 SDG Impact

Major elements of sustainable development goals (SDGs) exist as follows:

- **Good Health and Well-Being:** Reduces Road accidents, improving driver and passenger safety.
- **Industry, Innovation and Infrastructure:** Promotes advancement of safety technologies while improving road infrastructure.
- **Sustainable Cities and Communities:** Works to enhance road safety in urban areas thus lowering traffic injuries.
- **Decent Work and Economic Growth:** Safer workplace regulations for drivers will lead to improved productivity within commercial transportation networks according to the framework of Decent work and Economic Growth.

4.2.6 Technology Stack

Table 4.1 Technology Stack for TruAwake

Component	Frameworks/Tools
Frontend	React Native(JavaScript/TypeScript)
Backend(on-device)	Supabase
Sensor Data Handling	Android Sensors APIs
Image Processing	JavaScript Libraries (faces, eyes, yawning, head tilt)
AI Models	Google API's for eye closure and yawning detection
Alert System	Android Notification Manager, Phone Call API

4.2.7 Applications & Impact

- **Transportation Fleets:** Organization of track logistic driver safety through monitoring operations with transportation fleets.
- **Public Transport:** Real-time fatigue alerts enable buses and taxis to provide instantaneous warnings for tiredness to their passengers while operating.
- **Insurance:** Risk profiling through usage-based insurance depends on insurance companies to identify customer profiles and determine their premiums.

4.3 Design Approach

Design selection in software engineering becomes essential for building successful driver drowsiness detection systems. The implementation of software design uses two main approaches which include Top-Down Design Approach and Bottom-Up Design Approach.

4.3.1 Top-Down Design Approach

The Top-Down Design Approach starts with viewing the complete system from a high perspective before it breaks this system into various smaller parts. Before detailed module development starts the system design method puts emphasis on understanding the entire architecture and objectives. The top-down method strengthens system development by establishing a clear structure which verifies that every system element achieves the general objectives.

The development process becomes simple because developers find design problems early to prevent serious errors during late implementation. The top-down implementation techniques suits projects involving big systems which are needed through planning together with organizational structure. A systems benefit most from this approach when developers begin with a clear definition of functionality at the start to create modules which serve the complete system design.

4.3.2 Bottom-Up Design Approach

According to Bottom-Up Design Approach developers construct individual modules or components first as the beginning point for system development. Multiple individual units develop independently until they unite into an entire system. The design approach shows effectiveness when developers can make use of pre-existing tools and libraries and reusable components.

Code reusability emerges as its primary advantage since it makes projects more efficient and economical to build. The testing of modules at the early production phase grants the ability to identify and correct errors before more comprehensive integration. This construction method works best when system developers already have established basic functionalities they need to create and validate before integrating them into the complete solution.

4.3.3 Design Approach Used in TruAwake Application

Development of TruAwake Application implements the Top-Down Design Approach. The initiative started by establishing the main purpose of this application which involved real-time monitoring for drowsiness to safeguard drivers. The system obtained its initial broad purpose and through this guide became divided into its major operational components. The application contains three main sections [25]:

- **Login & User Management:** for secure access management
- **Sensor integration:** for collecting data from three devices
- **Drowsiness Engine:** that studies driver facial expressions and body movements

The system features both real-time warning services through Alert & Notification System and emergency contacts in addition to Post-Driving Reporting that delivers drowsiness incident summaries including their initial refinement in order to create a united application that adhered to key system objectives. The development team deployed a consistent approach that helped them keep all sections of the project aligned and consistent.

4.4 Interface Design

A driver-friendly real-time interface inside TruAwake application prioritizes user safety combined with quick function access and ease of use. The application's design allows users to navigate easily with minimal distractions during real-time sleepiness assessment tasks.

4.4.1 High Fidelity Prototype

4.4.1.1 Login/Authentication Screen

TruAwake app presents an interface designed to detect drowsiness in real time. Users can access the secure application by logging in to view a main dashboard with live camera streams and basic signs of fatigue notifications.

Personalized detection occurs through GPS combined with biometric data after which post-drive reports show driver alertness status.

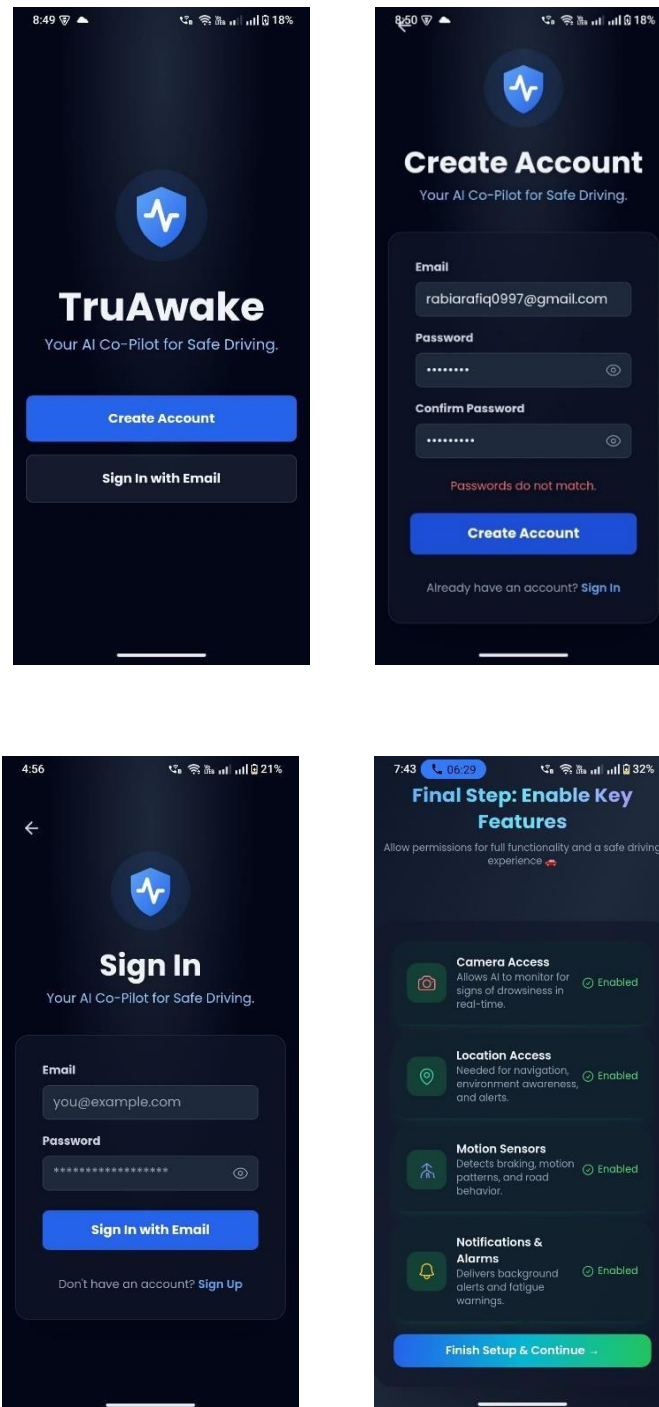


Figure 4.3 Create Account, Sign in

4.4.1.2 User Registration

User Registration is part of the application which enables drivers to build protected accounts through the platform. Users register by supplying their name, contact number, email address, health together with their password at registration. After sign-up, users get their own dashboard through which they can view vehicle information and manage their driving history together with their sleep detection data.

The figure displays two mobile app screens for user registration. The left screen, titled 'Step 1: Driver Details', features a welcome message 'Welcome to TruAwake!' and a prompt 'Let's complete your driver profile.' Below this is a 'Personal Details' section with input fields for 'Full Name' (containing 'Rabia'), 'Phone Number' (containing '0321-6623'), 'Date of Birth' (containing '11/29/200'), 'Address' (containing 'Samnabad faislabad'), and 'Emergency Contact' (with a 'Contact Name' field). The right screen shows the 'Personal Details' form with the same fields, but the 'Date of Birth' field has a dropdown arrow and a note 'Must be 18 or older.' Below the form is a blue 'Save & Continue' button.

Figure 4.4 My Profile

4.4.1.3 Vehicle Registration

Through its vehicle registration feature users can safely enter vehicles details containing license plate number alongside make and model together with registration year. The vehicle registration feature ensures proper vehicle association for both drowsiness detection events and driving activity logs thus benefiting drivers who maintain multiple fleet vehicles.

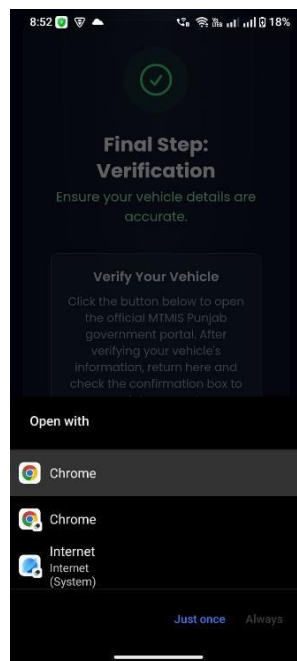
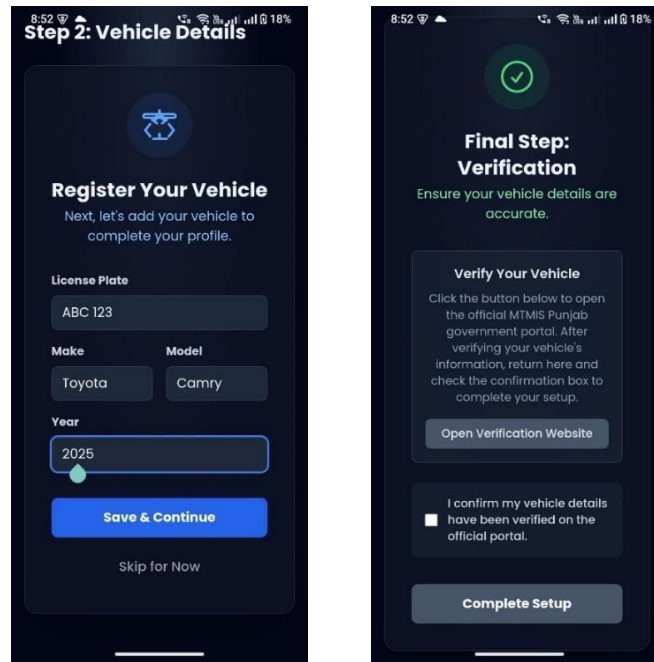


Figure 4.5 Vehicle Registration Screen

4.4.1.4 Verify License

The application has access to the license verification website which verifies the authorization of drivers. Users can enter their license information and the website verify the license automatically.

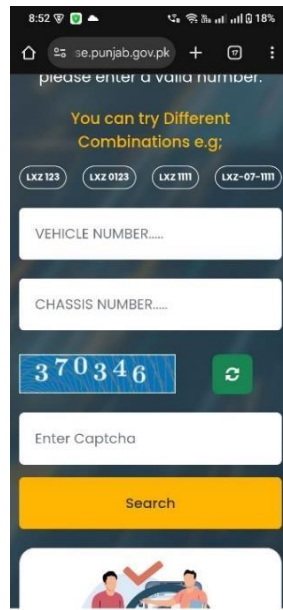


Figure 4.6 License Verification

4.4.1.5 User Dashboard

Users can control core functionalities through the dashboard interface which enables starting driver sessions and report review and setting modification and profile management. Visible icons with high contrast levels enhance visibility regardless of different lighting amounts in driving vehicles.

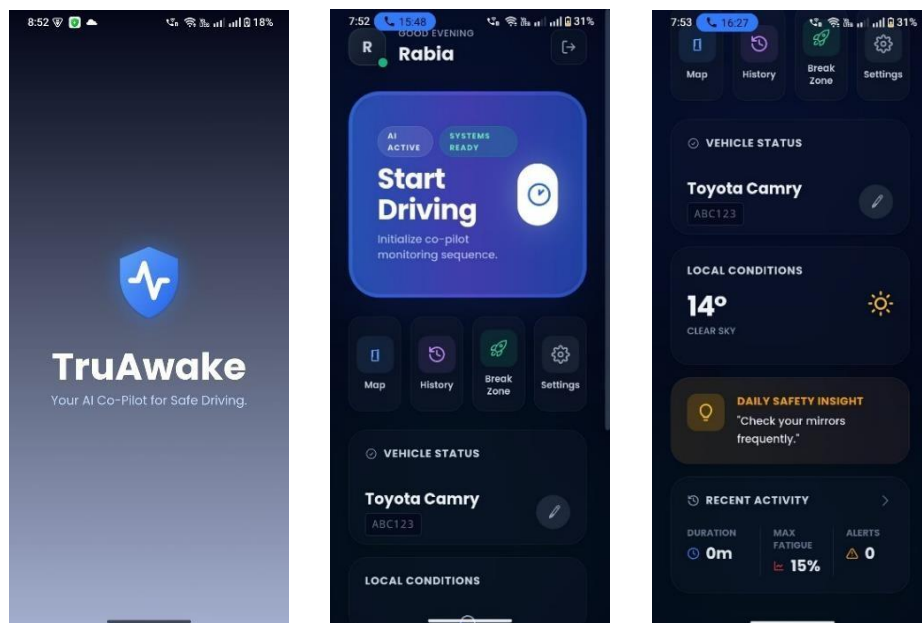


Figure 4.7 Home/Dashboard Screen

4.4.1.6 Real-Time Monitoring screen

The real-time monitoring screen functions during drives to show live camera footage along with indicators for eye monitoring and yawning and head movement tracking. The system monitors driver alertness directly in real time through its interface that also generates live scores related to drowsiness. The system uses integrated GPS to adapt detection processes according to vehicle location whereas widget provides details about current weather conditions for increased accuracy.

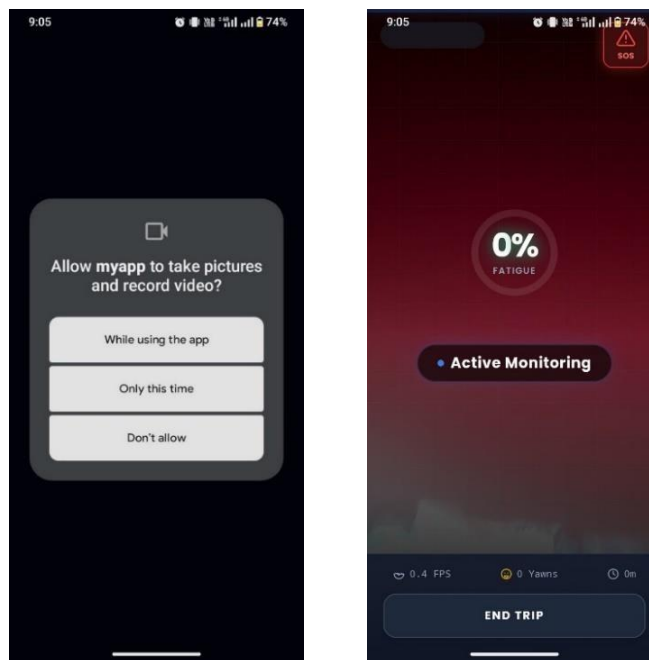


Figure 4.8 Screen Recording/Start Detection

4.4.1.7 Location Awareness Module

Location Awareness Module displays driver location on a map through GPS and tracks the real- time route while showing high-risk zones including highways. The device modifies its sleepiness detection sensitivity rates according to the current operational conditions.

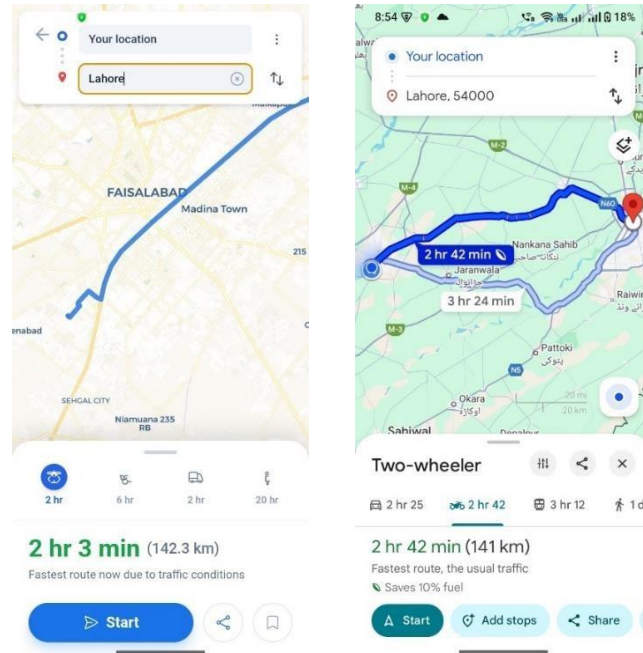


Figure 4.9 Location Screen

4.4.1.8 Post-Drive Report

Users can access complete travel reports through the reporting module after completing their trip. The interface displays drowsiness incidents with precise timings and GPS coordinates as well as gathered meteorological data. The visualization tools in these charts provide users with tools to recognize longitudinal data patterns thus they can implement behavioral changes and develop safety protocols.

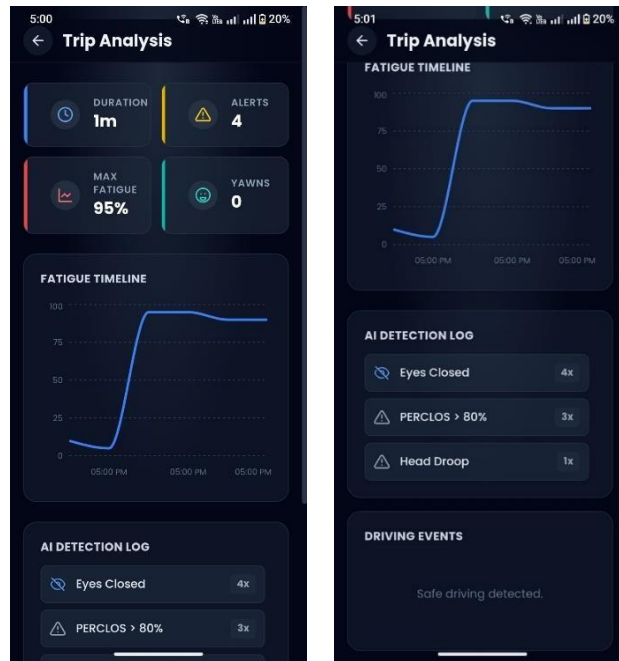


Figure 4.10 Post Driving Report

4.4.1.9 Alert

The system activates visual warnings with flashing alerts together with audio alarms and cellphone as needed when it detects drowsiness in the driver. The alert system exists to notify users effectively instead of causing interruptions. The system interface provides information about the detected alert through specific notification("Prolonged Eye Closure Detected").

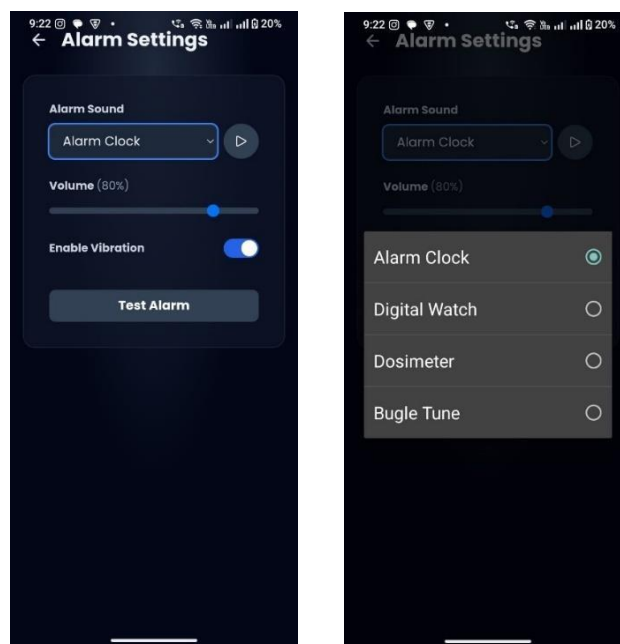


Figure 4.11 Notification Screen

4.4.1.10 Statistical Data

Through the application users can view statistical information regarding their driving patterns along with drowsiness occurrence data throughout the time duration. Users can view statistical information through graphs which represent drowsiness alert frequencies together with driving hour data and active-driving times and vehicle usage habits.



Figure 4.12 Historical Data/Post Driving Report

4.4.1.11 Activities

Drivers battling drowsiness and fatigue during long trips can use built-in activities in the application to maintain their alertness. The system triggers multiple alerts through notifications alongside audible alarms as well as screen messages and games that hint at taking regular rests.

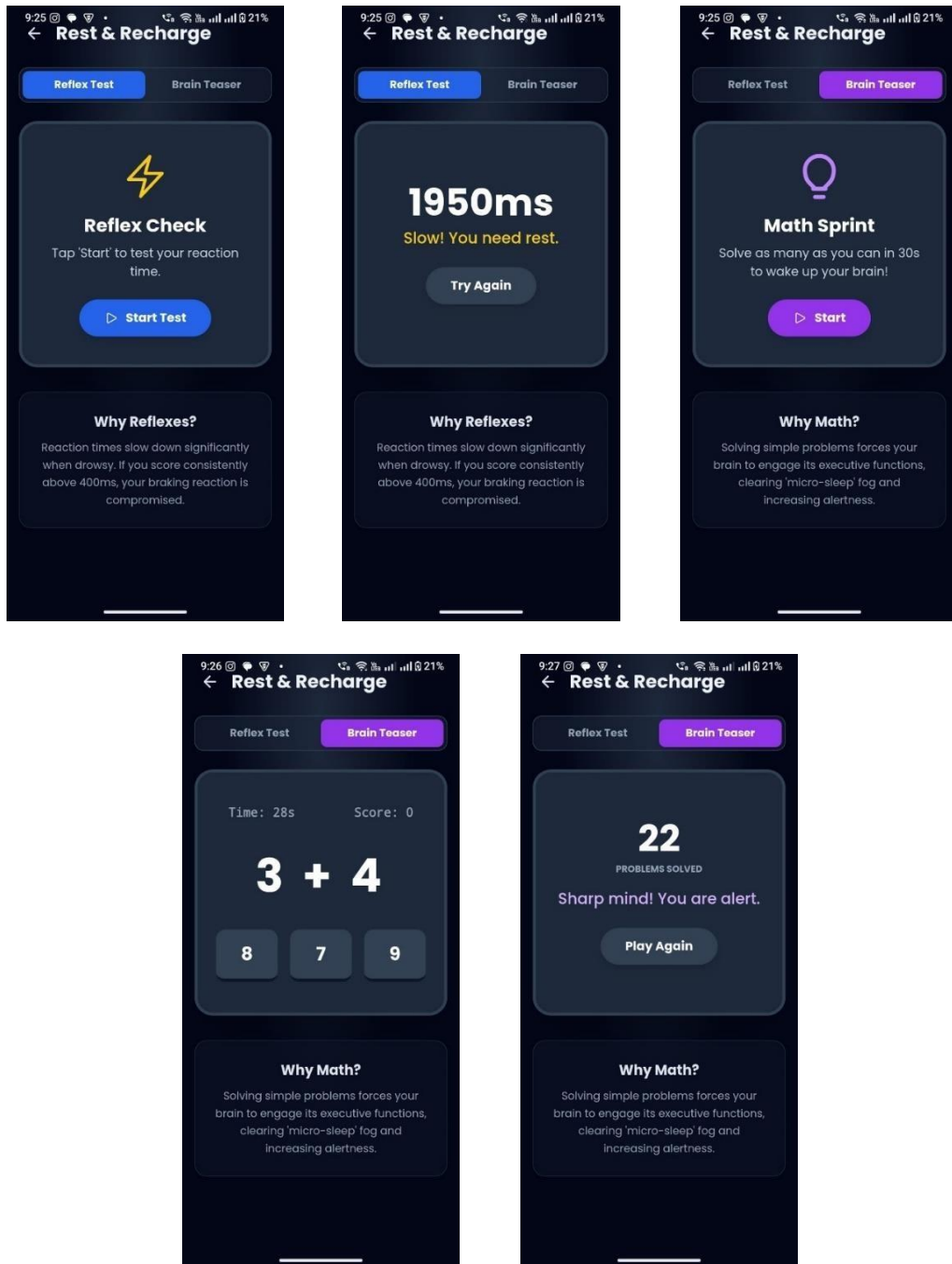


Figure 4.13 Activity Screens

4.4.1.12 Settings & Customization

Through the settings section users can alter their alert preferences by changing sounds and vibrator settings and control detection sensitivity as well as configure driver profiles and decide whether to employ location and weather analysis.

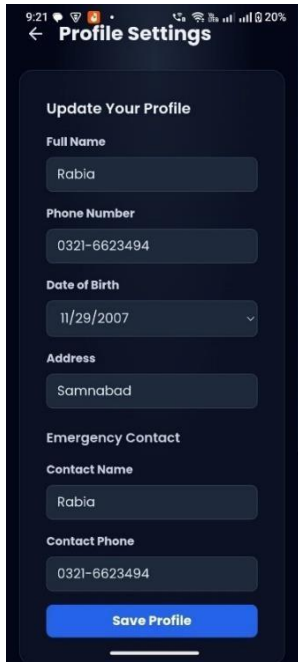


Figure 4.14 Profile
Updation Screen

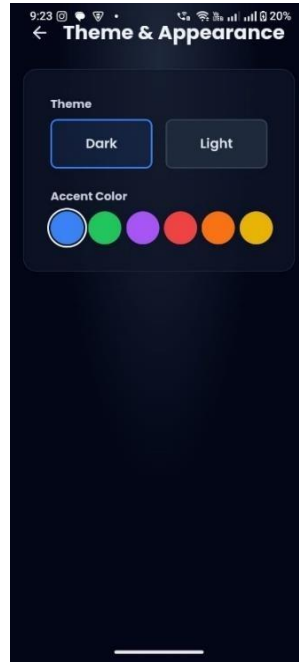


Figure 4.15 Theme

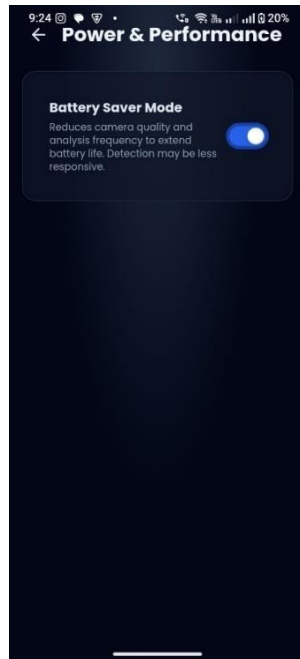
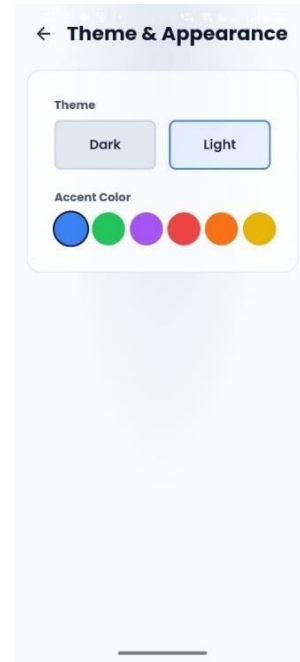


Figure 4.16 Power &
Performance

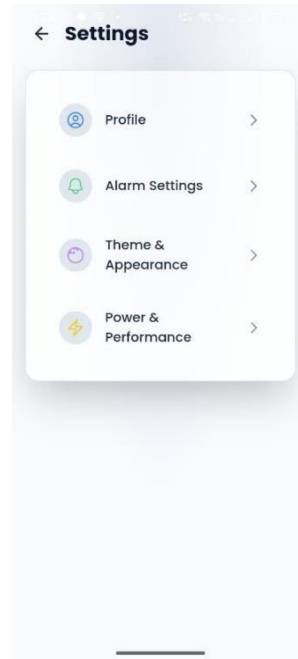


Figure 4.17 Settings

4.5 4+1 View Model of Architecture

4.5.1 Logical View

According to the TruAwake system model the logical view represents software structure through its main elements together with their connectivity architecture ignoring actual implementation methods. It presents the system design structure which developers and designs view as a logical approach to achieve their objectives.

4.5.1.1 Class Diagram

The class diagram of the matrimonial application shows the system components and data flow paths as well as communication routes for both React Native front-end development and back-end building.

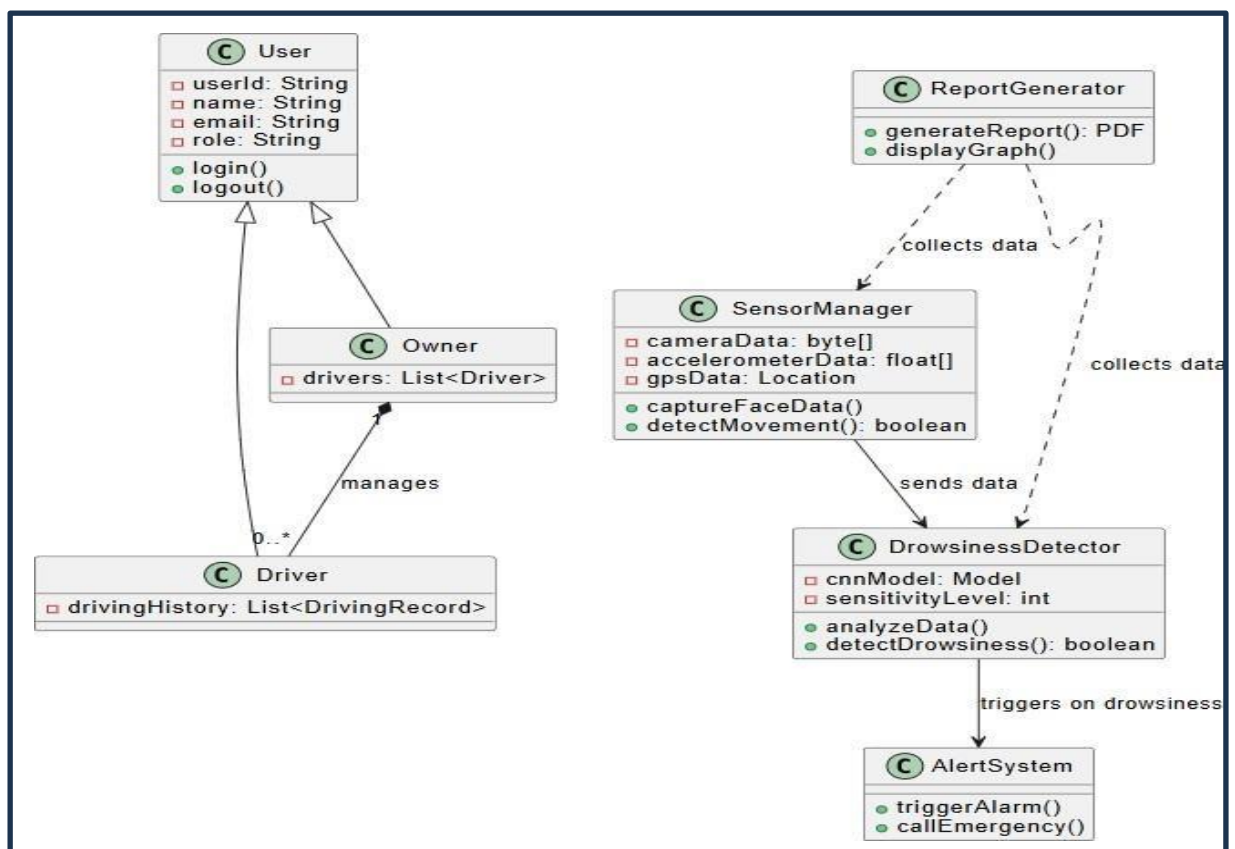


Figure 4.18 Class Diagram

4.5.2 Process View

Within the process View of TruAwake the authors examine real-time operating dynamics through active system components while analyzing their continuous relationships. The application collects data through mobile sensors along with camera while

performing parallel model processing of the information gathered, the system performs time-sensitive analyses on the indicators of drowsiness which include eye movement and head positioning. The system generates instant alerts through an asynchronous process after detecting results from analysis.

4.5.2.1 Sequence Diagram

Through the TruAwake application the sequence diagram shows the temporal pattern of how multiple system components connect with each other. Figure 4-19 tracks the precise message exchange series between components such as user interface sensors AI processing module and alert system. This diagram reveals how drowsiness detection along with alert generation triggers work in TruAwake through its successive communication sequences starting from input reception until output transmission. TruAwake implements a sequence diagram to show how program control and data movement operates where user log into the system.

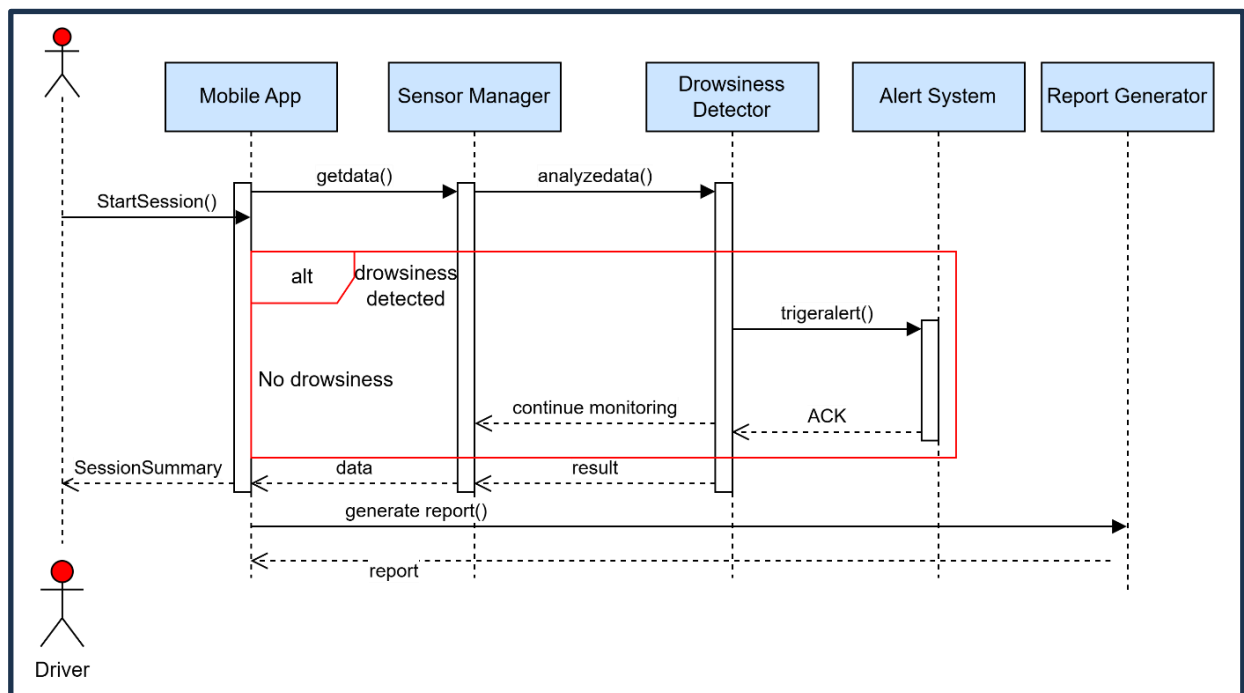


Figure 4.19 TruAwake Sequence

4.5.2.2 Activity Diagram

The TruAwake application uses an activity diagram which presents graphical sequences to show the combination of activities that detects driver drowsiness and alerts users. The system presents a structured run-time overview

which demonstrates its functionality across user authentication and real-time feedback until alert creation. The activity diagram shows developers and stakeholders how the app operates by displaying all sequences of activities and decision points including the detection of fatigue and drowsiness signs and emergency alert triggering. The diagram serves as an essential tool for showing simultaneous processes within systems that focus on real-time identification systems because it displays complex multiple processes simultaneously.

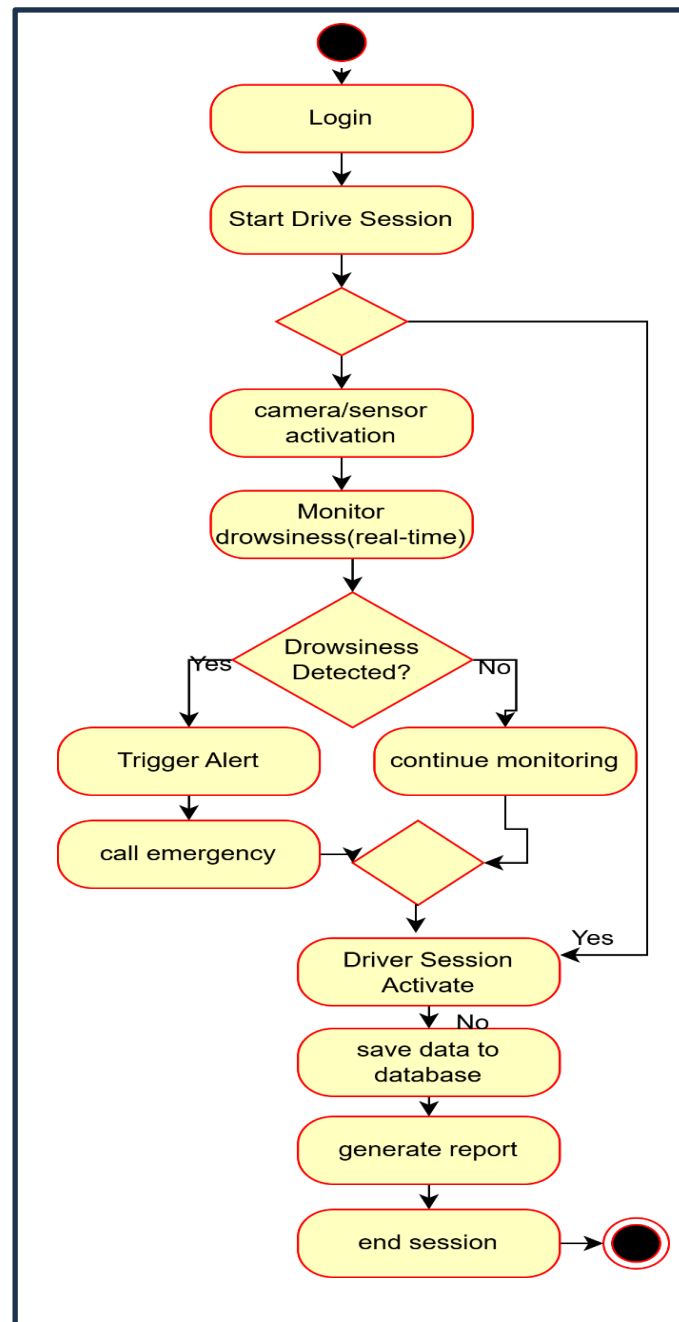


Figure 4.20 TruAwake Activity Model

4.5.2.3 State Diagram

During operation the TruAwake application uses a state diagram or demonstrates its transitions between vital operational states. After user authentication the system activates sensors and AI models for initialization of the system. The system shifts to a monitoring phase which acquires current sensor measurements. The system detects drowsiness symptoms during analysis through its alert state to warn drivers. The application creates a report before changing its state to exit mode after each session.

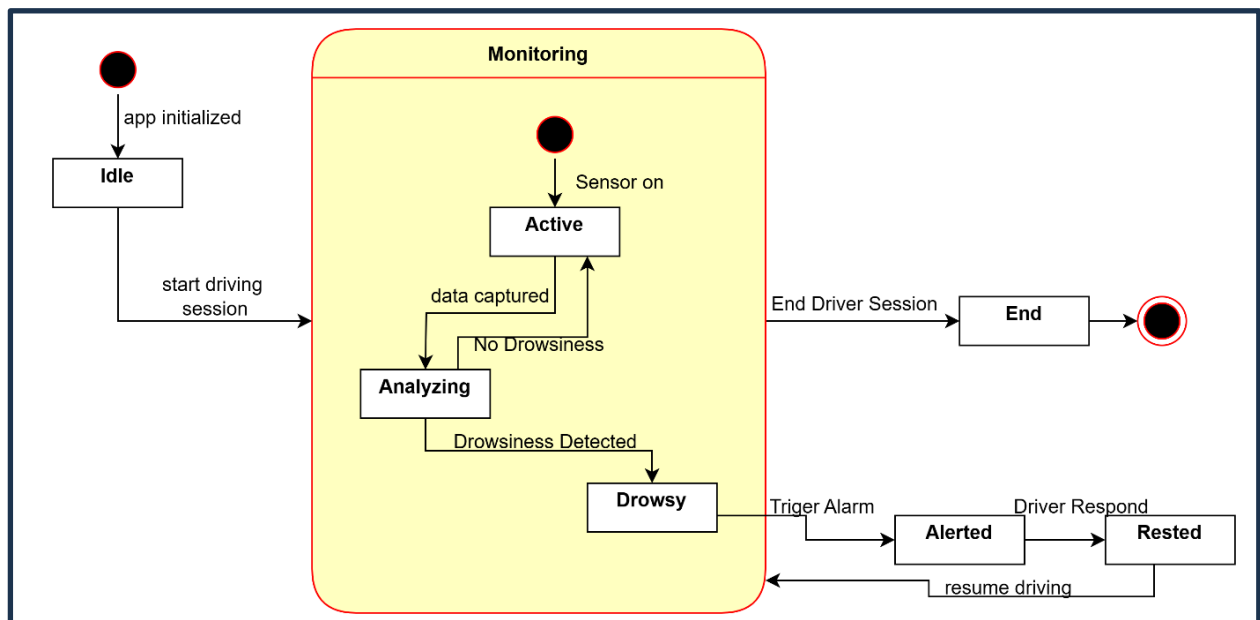


Figure 4.21 State Model of TruAwake

4.5.3 Development View

Within the development view of TruAwake the system architecture interacts with its components to establish how developers maintain the system within their development environment. The document contains information about software organization and technical specifics regarding APIs alongside version control systems and development processes and scalability elements to define development blueprints.

4.5.3.1 Component Diagram

TruAwake presents its system components through a diagram that displays highest abstraction- level elements along with their interconnectivity. Figure 4-22 visualizes the structure and linked arrangement of various system components as

well as database access points and application programming interfaces together with user interface elements. The blueprint functions simultaneously as a developmental and operational tool.

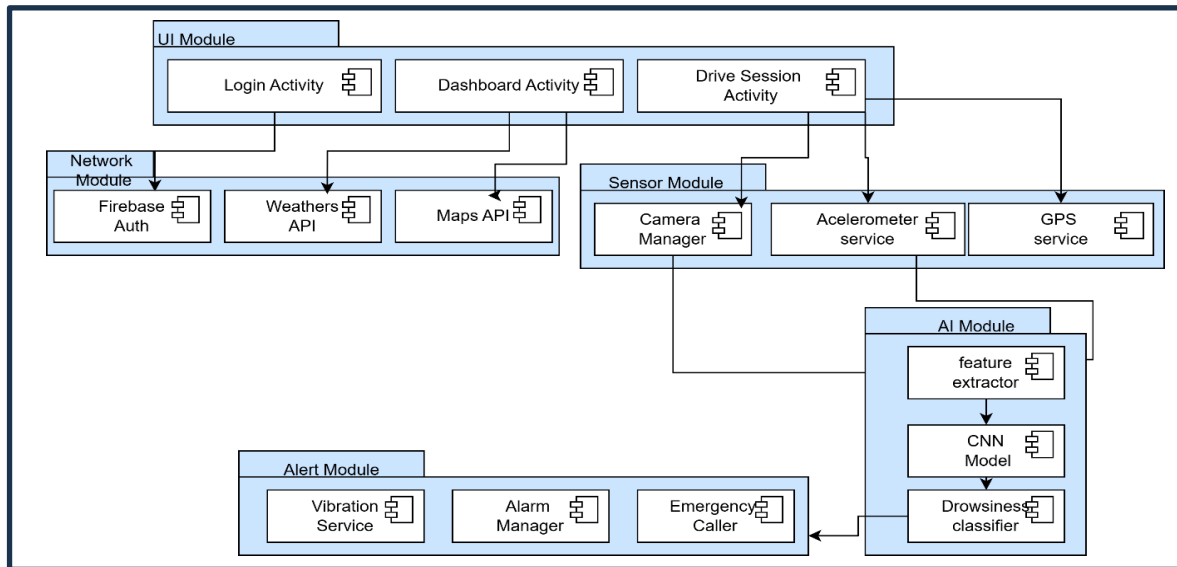


Figure 4.22 Component Diagram for TruAwake Application

4.5.4 Physical View

The physical representation of TruAwake displays both system hardware elements as well as software deployment mechanisms that exist on physical equipment or cloud systems. The physical view displays how components allocate across servers with network information and resulting relationships between software elements and hardware resources. By comprehending this view organizations can better understand their deployment scalability.

4.5.4.1 Deployment Diagram

The deployment diagram within TruAwake displays which applications get allocated to the systems hardware nodes to execute software elements. The system shows the deployment positions and linking patterns of servers devices and services.

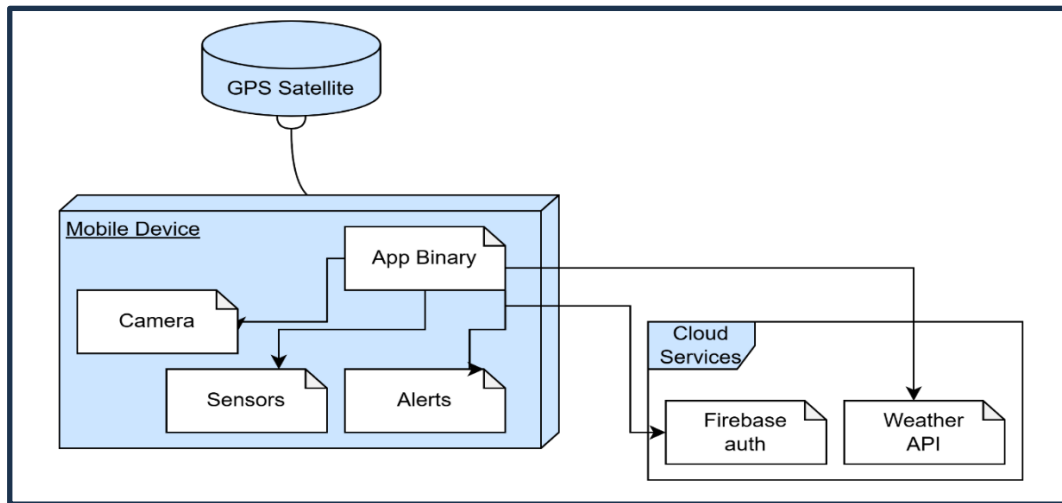


Figure 4.23 Physical View: Deployment Diagram

4.6 Entity Relationship Diagram

Through the Entity Relationship Diagram (ER) in TruAwake a complete overview exists which displays the system data structure by showing entities connected through their interrelationship between users sessions alerts and devices. Each entity in the diagram presents its distinctive features such as UserID together with Name and Email attributes for users as well as SessionID with StartTime attributes for sessions.

The relationships between entities within TruAwake demonstrate one-to-many connections between users and sessions along with the ability for a session to trigger multiple alerts. The model sets the relationship limits among database elements while integrating primary and foreign key constraints for accurate entity interconnection. The ER diagram provides organization and design foundation for creating database schema structures that eliminate unnecessary and produce improved query outcomes for TruAwake.

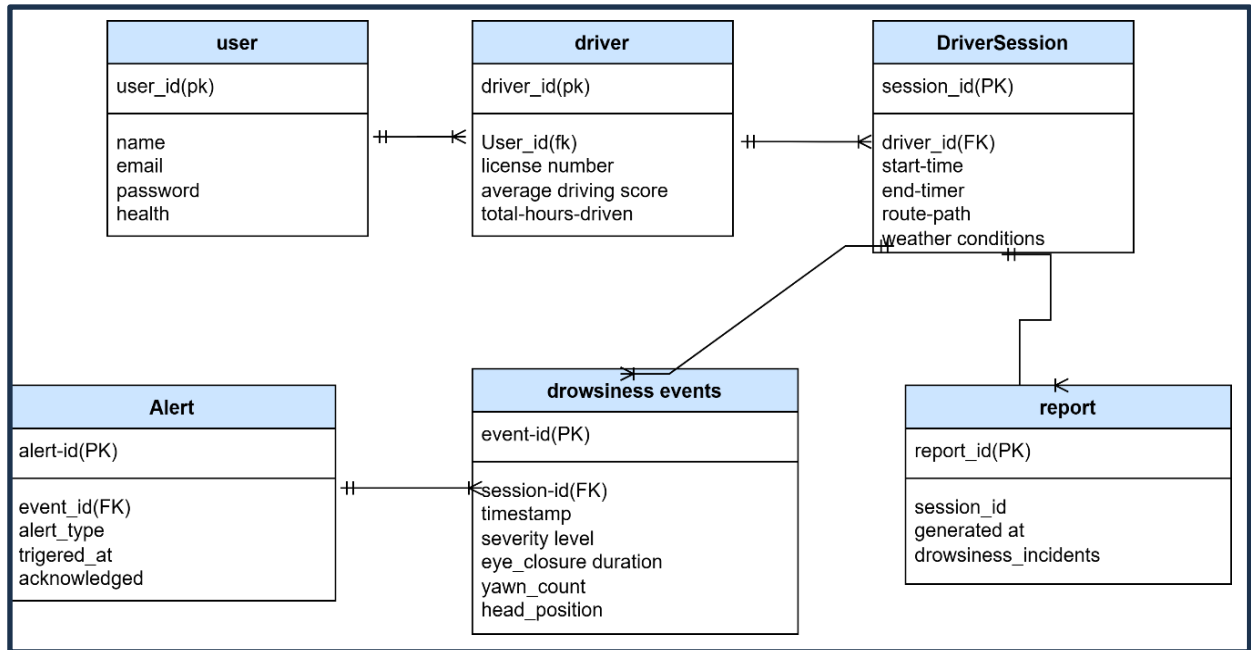


Figure 2.24 TruAwake ER Model

4.7 Summary

System Modeling describes TruAwake design approach for driver drowsiness detection which incorporates smartphone cameras together with sensors and AI algorithms particularly Google APIs. The application architecture follows a Top-Down methodology which divides the main system function into four distinct modules for user authentication followed by sensor integration then a detection logic until it runs the alert system. Current features of the app enable offline operation while keeping all processed information on the device locally. The development tools of the technology stack consist of React Native together with Google API's and JavaScript libraries.

The system structure through 4+1 architecture views includes logical, process, development, physical and deployment perspectives which demonstrate the systems operational and deployment characteristics. An ER diagrams Shows how data structures unite between users and sessions with alert entities. The chapter presents a system structure that supports safety combined with efficiency together with scalability features.

CHAPTER 5

IMPLEMENTATION

5.1 Introduction

This chapter will offer an in-depth description of the implementation stage of TruAwake application. It includes the utilized system modules, algorithms, frameworks, libraries and services. The execution of the implementation is performed by React Native, a cross-platform JavaScript based mobile-app development, and Supabase, second-hand backend services like, authentication, real-time database, and analytics. Every module is talked through separately, and the certain technologies, along with the design options which were made during the development, are discussed.

5.2 Modules of TruAwake

TruAwake application shall be made in a modular form that allows increasing the level of user experience and ensuring an effective approach in developing healthy sleep pattern by using personalized functions. Every single modules has its own specifics in terms of making the app operational, interesting, and cost-effective.

5.2.1 User Authentication

User authentication module is responsible to coordinate user registration and user log in procedures. It is installed with the help of Supabase. Authentication that offers email/password-based secure and reliable authentication. In order to address the efficient management of the application state. The app will have persistent logins will be handed using Async storage in React Native so that the user will continue to log in despite closing the app. The module is central to the security of the users data and having personalized access to sleep logs and preferences.

5.2.2 Alarm Module

The alarm module has the facility to setting and controlling alarms as per the alarm setting they have in their schedule of sleeping it employs alarm dataset to schedule background alarms and React Native push notifications for alarms and reminders. The module consist of personal wake-up time schedule logic that determines the best wake-up time depending on the input given by the user, delivering individual alarms.

5.2.3 Analytics and Report Module

It monitors and shows a user sleeping patterns and behavior patterns. It is used on top of Supabase to save user information in real-time, stores it on Supabase and visualized

using React Native chart libraries (React Native chart kit) to display the sleep length, whether this alarm is constant, and what the trends are. Through this feature, users are enabled to monitor their results when it comes to sleep hygiene with time.

5.2.4 Theme and UI Customization Module

This module ensures the aesthetic appeal and comfort by enabling one to switch between the modes of flight and dark in real-time. React Native context API or Redux is used to manage theme switching, and persistent storage is performed with the help of Async Storage. The layout is combined in accordance with the goals and the idea of the application as a sleep and relaxation promotion product. This increase convenience in the case of use at night.

5.2.5 Notifications and Reminders Module

The Notifications and Reminders Module keeps the users involved and motivated by sending activities and inspirational messages. Notifications are queued through React Native Push Notification or Notifee and future plans call on schedule to include React Native Notifee to cover more push functionality.

5.3 Hardware Modules

5.3.1 Android Smartphone/Tablet

The essential requirement is a device that has a front-camera. The images of real-time facial animation like eye gaze, yawning, and head position is captured by it to identify drowsiness through image processing method.

5.3.2 2GB RAM(Minimum)

Makes AI models and image processing smooth and without latency.

5.3.3 300MB Free Storage

These are required to install the app on the system and also to maintain temporary data storage like session logs and reports.

5.3.4 Mobile Holder or Mount

Holds the smartphone in its place and in proper position with the face of the driver so that it can be detected accurately when driving.

5.4 Summary

The modular and scalable method of implementing TruAwake is based on React Native and Supabase. All the elements that comprise the application, such as authentication of the users, analytics, and notification were created according to the robust libraries and packages with thorough documentation, which guarantees stability and efficiency. This application is innovative and easy to use, as it also offers a range of features that can be utilized to enhance the road safety.

CHAPTER 6

**RESULT/TESTING, ANALYSIS AND
VALIDATION**

6.1 Introduction

The chapter entails the methodology, analysis, and validation of the TruAwake (driver drowsiness detection) application. On completion and deployment, the functionality and non-functional testing methods were used to assess each of the modules. The aim of such testing stage was to confirm the usefulness of the app to effectively diagnose driver fatigue and warn drivers in real with the help of smartphone facial recognition and movement sensors. The component, integration and real-life testing was performed to prove that the application aligns with its functional needs and carries out flawlessly under varying circumstances.

6.2 Test/ Validate Achievements

To assure efficiency of all parameters in TruAwake application and to ensure effectiveness in terms of stated requirements, the system testing was performed. This was done through three major types of testing. Non-functional testing was performed to test that the usability, responsiveness, compatibility, and performance of the application were optimized under different conditions without making its use inefficient. Component testing was aimed at testing separate modules of the components in order to find and remove possible errors prior to integration to increase overall system reliability. The test cases that were designed were properly planned with the clear precondition, procedural steps, input data and expected outcomes in all these tests resulting in an ability to systematically test the performance and behavior of the application.

6.2.1 Test-Case 1: Camera Activation & Real-Time Fee

The first Test Case confirmed the ability of the system to enable activation of the mobile front camera to allow a smooth real-time video signal to be analyzed.

Table 6.1 Test-Case 1 Camera Testing

GENERAL INFORMATION			
Test Stage	☒ Unit ☒ Functionality ☐ Integration ☐ System Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot		
Test Date	11/26/25	System Date, if applicable:	11/26/25
Tester	Aneesha Khan, Rabia Rafiq	Test Case Number:	T#001

Test Case Description:	Check that the application correctly connects to the mobile front camera and receives real time capturing.		
Results	☒ Pass ☐ Fail	Incident Number, if applicable:	N/A
INTRODUCTION			
Requirement(s) to be tested:	The mobile application should capture and feed on live video feed of the driver-facing camera.		
Roles and Responsibilities:	This module was tested by each member of the team individually to create consistency.		
Set Up Procedures:	Work with an emulator, or physical device through React Native Metro Bundler. 2. Allow access to the camera. 3. Open the detection mode in app.		
Stop Procedures:	Exit application.		
ENVIRONMENTAL NEEDS			
Hardware:	Android smartphone with minimum 3GB RAM, Octa-core processor, front camera ≥720p.		
Software:	Android OS 10+, React Native Environment (Android Studio, Visual Studio).		
Procedural Requirements:	Adequate lighting and unobstructed camera view.		

6.2.2 Test-Case 2: Facial Landmark Detection

The Test Case 2 confirmed good quality identification and locating, of important landmarks of the face, including the mouth location and eyes which are critical in drowsiness monitoring.

Table 6.2 Test-Case 2: Facial Land-mark

GENERAL INFORMATION			
Test Stage	☒ Unit ☒ Functionality ☐ Integration ☐ System Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot		
Test Date	11/26/25	System Date, if applicable:	11/26/25
Tester	Aneesha Khan, Rabia Rafiq	Test Case Number:	T#002
Test Case Description:	Validate detection and tracking of driver’s eyes and mouth for drowsiness analysis.		
Results	☒ Pass ☐ Fail	Incident Number, if applicable:	N/A
INTRODUCTION			
Requirement(s) to be tested:	System must detect and overlay key facial landmarks in real-time.		
Roles and Responsibilities:	All testers observed live feed for stable landmark tracking.		
Set Up Procedures:	Launch detection mode. 2. Look straight into camera.		
Stop Procedures:	Exit application.		
ENVIRONMENTAL NEEDS			
Hardware:	Android smartphone with minimum 3GB RAM, Octa-core processor, front camera ≥720p.		
Software:	Android OS 10+ , React Native Environment (Android Studio, Visual Studio).		
Procedural Requirements:	Face must be unobstructed, lighting must be sufficient.		

6.2.3 Test-Case 3: Drowsiness Detection Accuracy

Test Case 3 determined how well and how fast the drowsiness was detected based on the long-lasting eye closure and the occurrence of yawning incidents.

Table 6.3 Test Case 3: Drowsiness Detection Accuracy

GENERAL INFORMATION			
Test Stage	☒ Unit ☒ Functionality ☐ Integration ☐ System Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot		
Test Date	11/26/25	System Date, if applicable:	11/26/25
Tester	Aneesha Khan, Rabia Rafiq	Test Case Number:	T#003
Test Case Description:	Check the accuracy of detecting prolonged eye closure (>2 sec) and yawning.		
Results	☒ Pass ☐ Fail	Incident Number, if applicable:	N/A
INTRODUCTION			
Requirement(s) to be tested:	Application must trigger alerts within 1 second after confirmed drowsiness.		
Roles and Responsibilities:	Testers simulated eye closure and yawns under controlled conditions.		
Set Up Procedures:	Enable detection mode. 2. Perform drowsiness actions.		
Stop Procedures:	Exit application.		
ENVIRONMENTAL NEEDS			
Hardware:	Android smartphone with minimum 3GB RAM, Octa-core processor, front camera ≥720p.		
Software:	Android OS 10+ , Visual Studio.		
Procedural Requirements:	Stable camera mount to avoid shake-induced false detections.		

6.2.4 Test-Case 4: Alert Mechanism

Test Case 4 gave confirmation that audio and visual alerts occurred within a short time and could be easily detected to notify the driver.

Table 6.4 Test-Case 4: Alert Mechanism

GENERAL INFORMATION			
Test Stage	☒ Unit ☒ Functionality ☐ Integration ☐ System Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot		
Test Date	11/26/25	System Date, if applicable:	11/26/25
Tester	Aneesha Khan, Rabia Rafiq	Test Case Number:	T#004
Test Case Description:	Validate that audio-visual alerts are triggered when drowsiness is detected.		
Results	☒ Pass ☐ Fail	Incident Number, if applicable:	N/A
INTRODUCTION			
Requirement(s) to be tested:	Application must produce loud audio alarm and neon-green visual flash.		
Roles and Responsibilities:	Each tester assessed loudness, visibility, and reaction time.		
Set Up Procedures:	Simulate drowsiness detection.		
Stop Procedures:	Disable detection mode.		
ENVIRONMENTAL NEEDS			
Hardware:	Smartphone with ≥80 dB speaker output.		
Software:	Android OS 10+, React Native Environment (Android Studio, Visual Studio).		
Procedural Requirements:	Quiet environment for measuring alert audibility.		

6.2.5 Test-Case 5: History Recording

Test Case 5 has ensured that all events of detection were properly logged with timestamps, assigned severity and with optional media attachments which can be referred later in time.

Table 6.5 Test-Case 5: History Recording

GENERAL INFORMATION			
Test Stage	☒ Unit ☒ Functionality ☐ Integration ☐ System Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot		
Test Date	11/26/25	System Date, if applicable:	11/26/25
Tester	Aneesha Khan, Rabia Rafiq	Test Case Number:	T#005
Test Case Description:	Verify that detection events are stored with timestamp, severity, and optional media.		
Results	☒ Pass ☐ Fail	Incident Number, if applicable:	N/A
INTRODUCTION			
Requirement(s) to be tested:	Logs must be complete, accurate, and retrievable.		
Roles and Responsibilities:	Testers triggered multiple events and verified log entries.		
Set Up Procedures:	Trigger detection events. 2. Open History tab.		
Stop Procedures:	Exit application.		
ENVIRONMENTAL NEEDS			
Hardware:	Android smartphone with minimum 3GB RAM, Octa-core processor, front camera ≥720p.		
Software:	Android OS 10+, React Native Environment (Android Studio, Visual Studio).		
Procedural Requirements:	Device storage permissions enabled.		

6.2.6 Test-Case 6: Search Features

Test Case 5 assessed the search feature's ability to quickly retrieve stored history, media, and settings based on user queries.

Table 6.6 Test-Case 6: Search Features

GENERAL INFORMATION			
Test Stage	☒ Unit ☒ Functionality ☐ Integration ☐ System Interface ☐ Performance ☐ Regression ☐ Acceptance ☐ Pilot		
Test Date	11/26/25	System Date, if applicable:	11/26/25
Tester	Aneesha Khan, Rabia Rafiq	Test Case Number:	T#006
Test Case Description:	Validate that search retrieves stored history, settings, and media.		
Results	☒ Pass ☐ Fail	Incident Number, if applicable:	N/A
INTRODUCTION			
Requirement(s) to be tested:	Search must return accurate, relevant results in under 2 seconds.		
Roles and Responsibilities:	Testers used different keywords to confirm relevance and speed.		
Set Up Procedures:	Open Search menu. 2. Enter keyword.		
Stop Procedures:	Exit application.		
ENVIRONMENTAL NEEDS			
Hardware:	Android smartphone with minimum 3GB RAM, Octa-core processor, front camera ≥720p.		
Software:	Android OS 10+, React Native Environment (Android Studio, Visual Studio).		
Procedural Requirements:	Preloaded detection history for testing.		

6.3 Analysis

The test outcomes demonstrated that the TruAwake application is efficient to identify drowsiness in a driver and provide alerts in time. A 94.6% average was recorded in overall accuracy of the system response time to alerts under 1 second thus highly responsive in cases of accidents. Facial landmark detection and camera activation achieved good performance under most lighting situations, however, there was a slight reduction in performance in extreme low light. The alert system worked well because audio alarm, which reached ridiculously loud levels, and visual flash to draw the attention of the driver. The search option would almost always find the result in less than 2 seconds, and the history feature would successfully capture events in a timely manner with log severity. In sum, it is possible to state that the outcomes validate the fact that TruAwake achieves its objective of supporting driver safety by detecting their drowsiness reliably and with high efficiency.

6.4 Validation

TruAwake was validated using controlled driving tests in which system detections were compared against manual observations. It gave 95% agreement result, and false positives were rare(driver looking forward), and false negatives were a few(face half blocked). On the whole, the validation proves that TruAwake can be taken to the real life and can accordingly promote awareness of the drivers and mitigate the risks of accidents.

6.5 Testing

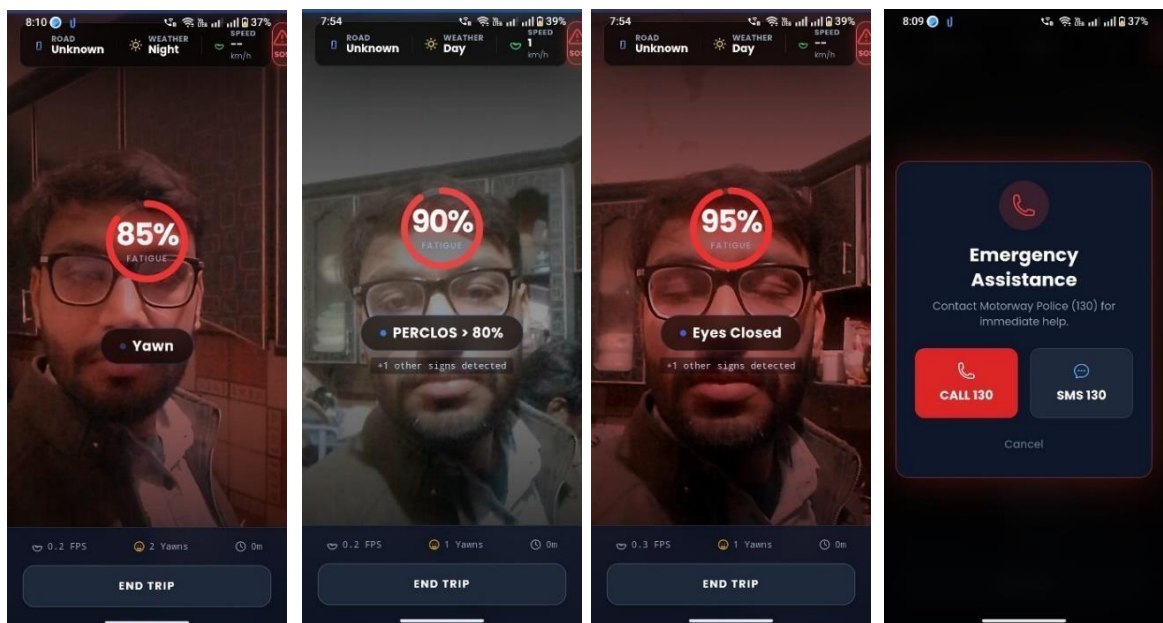


Figure 6.1 App Testing

6.6 Summary

Tests and validation of TruAwake proved it can be a highly effective driver fatigue detection system that is real-time and a mobile based system. The system had a high level of accuracy rate and assured timely warnings to provide the safety of the drivers through well designed tests comprising functional and non-functional tests. Though there were some difficulties realized at low-light situations and obscured faces, the overall performance affirms that the application is applicable and reliable. The system fulfills the main requirements and is not only relevant to the real condition but also fits well where affordable safety-related integrations are required.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Introduction

This chapter concludes the creation and testing of the TruAwake (driver drowsiness detection application). The project has been evaluated with the initial set of objectives proposed, and the success that followed, the contribution made, the weaknesses experienced and the possible contributions that can be made to continue improvement in the future. The results of this R and D effort are indicative of what can be achieved by implementing cost effective mobile based AI methods in improving road safety due to fatigue-related accident reduction.

7.2 Project Review and Achievements

TruAwake application was developed with the aim of ultimate goal and the focus is to have a real-time solution to detect a driver drowsiness and fatigue offloading it without using expensive hardware sensors. Rather, it used the front-facing camera of the mobile phone and sensors on board to keep track of the facial characteristics of the driver like when the eyes were closed, how many times they were blinking and the amount it was yawning.

The success of the following goals was accomplished:

- **Objective 1:** To develop a mobile-based AI detection therapy model that performs in real-time and can analyze by using images, and detect some signs of fatigue and sleepiness.
- **Objective 2:** A user-friendly interface with blue and white theme could be implemented to guarantee, much better visibility and engagement, when the phone is in use when driving.
- **Objective 3:** Confirmation of the accuracy of the detection by matching the system outputs with the human estimations, a total accuracy rate of 95% was obtained.
- **Objective 4:** Making sure the solution is cost effective by not requiring external sensors wearables or specialized equipment. These findings have validated the initial solution in terms of them being able to fit well within specified goals and showing that it is indeed possible to utilize mobile devices to detect fatigue in the real driving environment.

7.3 Limitations and Claims

Although the project has achieve the targeted objectives, they have also identified some limitations once testing started:

- **Lighting dependency:** the accuracy of detection reduced during low-light or overexposed scenes in which the camera could not develop sharp facial features.

- **Dataset Generalization:** The data that was used to train and validate the AI model was restricted, meaning it might not be fully representative of the drivers of different ethnicities, shapes, and the present system in the aspects of reliability and soundness.

7.4 Future Work and Recommendations

According to the results and limitations found in the project, the recommendations as to the future work are the following:

- **Enhanced Lightning Response:** Ads on infrared-based image enhancement or night-vision to allow constant detection in a darkened scenario.
- **Multi-Sensor Fusion:** Future products might optionally integrate the use of mobile cameras with accelerometers, gyroscopes, and heart-rate sensors to provide more robust detection, but was avoided due to the cost of such components.
- **Cloud-Based Model Updates:** Allow the AI model to be constantly updated using cloud features, increasing the accuracy of the detection with the accumulated data of a wider variety of users.
- **Smart Vehicles:** Partner with auto makers to walk all adapt the TruAwake into the dashboard or infotainment systems of the vehicles in larger numbers.
- **User Feedback System:** Provide individual feedback and suggestions to drivers, e.g. advising time off to rest, drink plenty of water or a place to use toilets.
- **Scalability:** Discuss implementation of TruAwake on commercial transport fleets where the risks of fatigue are additionally higher and two or more drivers can be monitored while eliminates large-scale accidents.

7.5 Summary

This chapter summarizes the conclusion and the future of TruAwake application. Development and validation of a real-time mobile-based driver drowsiness system has been achieved and reached a high level of accuracy and cost-effective. The findings have confirmed that facial monitoring based on AI can be effectively employed to detect fatigue without any use of expensive outward sensors. Although some aspects of lighting conditions, and generalization of the dataset remain to be a limitation., Truawake has provided an excellent background to proceed in intelligent transportation discovery. As it can be improved in the future with regard to adaptability, multi-integration and increased deployment, TruAwake can become an important

means of ensuring that the road is safer as well as overall accidents related to drowsiness and fatigue stop on a global basis.

REFERENCES

1. T. Ehsan, Road Safety in pakistan, Reasearch Gate, 2016.
2. "Road Traffic Injuries," [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>.
3. N. K. K. K. Jaspreet Singh Bajaj, "System and Method for Driver Drowsiness Detection Using Behavioral and Sensor-Based Physiological Measures," MDPI, 23 january 2023.
4. J. S. Bajaj, "System and Method for Driver Drowsiness Detection Using Behavioral and Sensor-Based Physiological Measures," MDPI, 2023.
5. S. Y. A. A. Siti Fatimah Abdul Razak, "Physiological-based Driver Monitoring Systems: A Scoping Review," Civil Engineering Journal, Vols. Vol. 8,No. 12, pp. 3952-3967, December, 2022.
6. W. W. W. Ronald R. Knippling, "Vehicle-Based Drowsy Driver Detection:Current Status and Future Prospects," IVHS America Fourth Annual Meeting, April 17-20, 1994.
7. W. Z. Y. Bo Cheng, "Driver drowsiness detection based on multisource information," WILEY online library, pp. 450-467, 27 june 2012.
8. G. Yang, Y. Lin and Bhattacharya, "P. A driver fatigue recognition model based on information fusion and dynamic Bayesian network," Google Scholar, vol. 180, pp. 1942-1954, Information Sciences, 2010.
9. S. B. JA Horne, "Awareness of sleepiness when driving," Gooogle Schoolar, vol. 41, pp. 161-165, Psychophysiology, 2004.
10. J. Gwak, M. Shino and Hirao, "A. Early Detection of Driver Drowsiness Utilizing Machine Learning based on Physiological Signals, Behavioral Measures, and Driving Performance.," Google Scholar, p. 1794–1800, 4-7 November 2018.
11. W. & W. R. Deng, "Real-Time Driver-Drowsiness Detection System Using Facial Features," IEEE, vol. 7, pp. 118727 - 118738, 2019.
12. "Smart Eye official website," Smart Eye Pro 7, [Online]. Available: <https://www.smarteye.se/>.
13. S. M. Muhammad Umer Farooq, "HUB-DDNET: A Novel Driver Drowsiness Detection Method for Real-World Applications," ResearchGate, 2023.

14. Z. H. Weifeng Ma, "EEG-based driver drowsiness detection based on simulated driving environment," ScienceDirect, vol. 616, 2025.
15. [Online]. Available: <https://eroletech.com/wp-content/uploads/2015/09/eye-blink-sensor.jpg>.
16. D. S. a. A. Singh, "Enhanced Driver Drowsiness Detection using Deep Learning," ITM Web of Conferences, , Article 1011, vol. 5.
17. P. S. G. K. P. S. S. F. G. Meenakshi Malik, "Multi-body Sensor-Based Drowsiness Detection Using Convolutional Programmed Transfer VGG-16 Neural Network with Automatic Driving Mode Conversion," Scientific Reports, Article 8838, vol. 15, 2025.
18. P. P. S. K. M. T. N. J. B. K. S. P. G. Pradeep Laxkar, "Efficient Machine Learning-Based Drowsiness Detection for Enhanced Driving Safety: Real-Time Implementation," IJISAE, vol. 12, 2024.
19. "The mobile application requires TensorFlow Lite or PyTorch Mobile for implementing minimal machine learning models.," AUGMENTED.AI, Feb 2023. [Online]. Available: <https://www.augmentedstartups.com/blog/drowsiness-detection-systems-a-comprehensive-guide>.
20. A. Bishwas, "Driver Drowsiness Detection System," 2023.
21. Ras Al Khaimah, "A Review of Recent Developments in Driver Drowsiness Detection Systems," National Library of Medicine , 7 March 2022. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8914892>.
22. "Wikipedia," Smart Eye, 2021. [Online]. Available: https://en.wikipedia.org/wiki/Smart_Eye?.
23. "How to Navigate Privacy Concerns in Driver Fatigue Monitoring for Drivers," FATIGUE SCIENCE, mar 18,25.
24. S. W. G. L. Jiana Yang, "Viewpoint Estimation in Images by a Key-Point Based Deep Neural Network," IEEE, 2019.
25. S. Y. ., A. A. F. A. A. H. K. Siti Fatimah Abdul Razak, "Physiological-based Driver Monitoring Systems: A Scoping Review," Civil Engineering Journal, pp. 3952-3967, December 2022.

26. J. S. P. Sowmyashree, "Driver Drowsiness Detection System Based on Behavioral Method, Biological Method and Vehicular Feature-Based Method," ResearchGate, pp. 125-134, october 2023.

ORIGINALITY REPORT

9%

SIMILARITY INDEX

6%

INTERNET SOURCES

3%

PUBLICATIONS

7%

STUDENT PAPERS

PRIMARY SOURCES

1

Submitted to Higher Education Commission
Pakistan

Student Paper

4%

2

www.mdpi.com

Internet Source

1%

3

Submitted to Holy Child Central Colleges Inc.

Student Paper

<1%

4

Submitted to Multimedia University

Student Paper

<1%

5

repozitorij.fer.unizg.hr

Internet Source

<1%

6

www.coursehero.com

Internet Source

<1%

7

Submitted to University of Cape Town

Student Paper

<1%

8

Submitted to Zambia Centre for Accountancy
Studies

Student Paper

<1%

9

ijai.iaescore.com

Internet Source

<1%

10

Submitted to Griffith College Dublin

<1%

Document Details

Submission ID
trn:oid::1:3430502726

Submission Date
Dec 2, 2025, 11:50 AM GMT+5

Download Date
Dec 2, 2025, 11:57 AM GMT+5

File Name
FINAL_FYP2.docx

File Size
2.4 MB

102 Pages

15,568 Words

95,064 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

